MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

Which of the following will pair up to produce stationary wave?

(1) $Z_1 = A\cos(kx - \omega t)$

(2) $Z_2 = A\cos(kx + \omega t)$

(3) $Z_3 = A\cos(ky - \omega t)$

(3) $Z_4 = A\cos(kz + \omega t)$

(A) 1 and 2

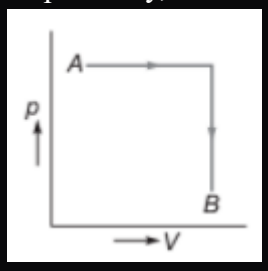
(B) 2 and 3

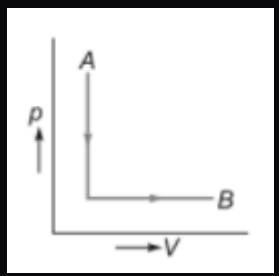
(C) 3 and 4

(D) 1 and 3

Q.2

In figure two indicator diagrams are shown. If the amounts of work done in the two cases are W_1 and W_2 respectively, then





(A) $W_1 = W_2$

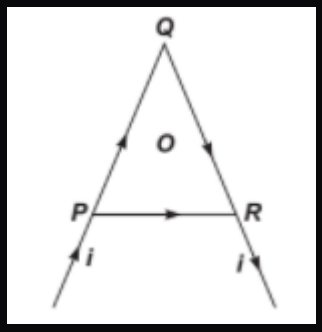
(B) $W_1 > W_2$

(C) $W_1 < W_2$

(D) Cannot say

Q.3

An equilateral triangle of side l is formed from a piece of wire of uniform resistance. The current i is fed as shown in the figure. The magnitude of the magnetic field at its centre O is



(A) $\frac{\sqrt{3}\mu_0 i}{2\pi l}$

(B) $\frac{3\sqrt{3}\mu_0 i}{2\pi l}$

(C) $\frac{\mu_0 i}{2\pi l}$

(D) zero

Q.4

A railway track is banked for a speed v, by making the height of the outer rail h higher than that of the inner rail. If the distance between the rails is l and the radius of curvature of the track is r, then

(A) $\frac{h}{l} = \frac{v^2}{rg}$

(B) $\tan\left\{\sin^{-1}\left(\frac{h}{l}\right)\right\} = \frac{v^2}{rg}$

(C) $\tan^{-1}\left(\frac{h}{l}\right) = \frac{v^2}{rg}$

(D) $\frac{h}{r} = \frac{v^2}{lg}$

Q.5

A double star system consists of two stars A and B which have time periods T_A and T_B. Radius R_A and $\mathrm{R_B}$ and mass $\mathrm{M_A}$ and $\mathrm{M_B}$. Choose the correct option.

(A) If $T_A > T_B$ then $R_A > R_B$

(B) If $T_A > T_B$ then $M_A > M_B$

(C) $\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{R_A}{R_B}\right)^3$

(D) $\mathrm{T_A} = \mathrm{T_B}$

Q.6

Two resistances of $400\ \Omega$ and $800\ \Omega$ are connected in series with 6 volt battery of negligible internal resistance. A voltmeter of resistance $10,000\ \Omega$ is used to measure the potential difference across $400\ \Omega$. The error in the measurement of potential difference in volt approximately is

(A) 0.01

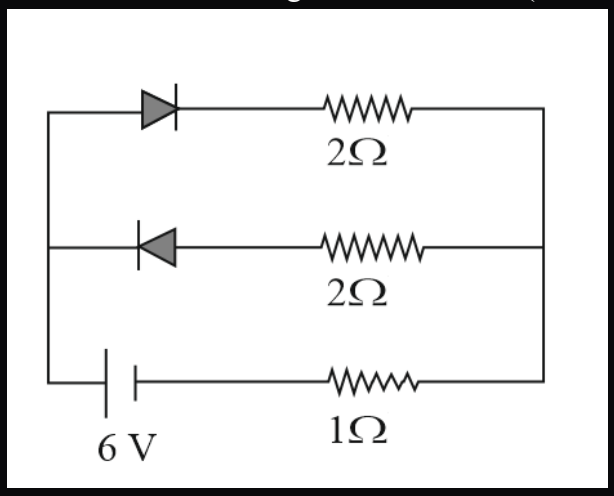
(B) 0.02

(C) 0.03

(D) 0.05

Q.7

Find the current through 1 Ω resistance. (Assume diode to be ideal)



(A) 2

(B) 3.5

(C) 3

(D) 2.5

Q.8

If the velocity of a particle is given by $v = (180 - 16x)^{1/2}\text{ms}^{-1}$, then its acceleration will be

(A) 0

(B) 8ms^{-2}

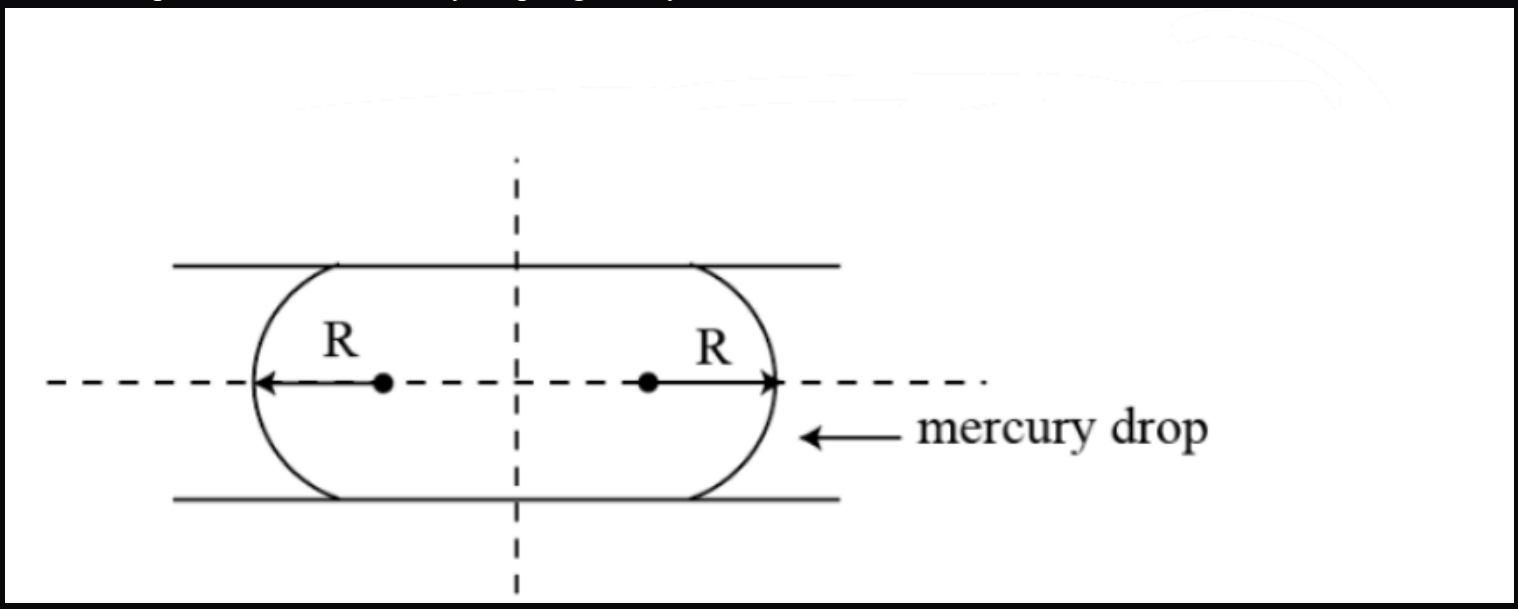
(C) -8ms^{-2}

(D) 4ms^{-2}

Q.9

A mercury drop lies between two glass plates separated by a very small distance (see figure). Surface tension of mercury is T. Radius of curvature of drop surfaces in a direction parallel to plates is R. The other surfaces are flat.

The excess pressure for the mercury drop is given by:



(A) $\frac{T}{R}$

(B) $\frac{2T}{R}$

(C) $\frac{3T}{R}$

(D) $\frac{4T}{R}$

Q.10

In an electromagnetic wave in free space the root mean square value of the electric field is $\mathrm{E_{rms}} = 6\ \mathrm{V/m}$. The peak value of the magnetic field is :-

(A) $2.83 \times 10^{-8}\ \mathrm{T}$

(B) $0.70 \times 10^{-8}\ \mathrm{T}$

(C) $4.23 \times 10^{-8}\ \mathrm{T}$

(D) $1.41 \times 10^{-8}\ \mathrm{T}$

Q.11

Two coils have a mutual induction $0.005\ \mathrm{H}$. The current changes in the first coil according to the equation $i = i_m \sin \omega t$ where $i_m = 10\ \mathrm{A}$ and $\omega = 100\pi\ \mathrm{rads}^{-1}$. The maximum value of the emf induced in the second coil is :

(A) $5\pi V$

(B) $4\pi V$

(C) $2\pi V$

(D) πV

Q.12

A particle of mass M at rest decays into two masses m_1 and m_2 with non-zero velocities. The ratio of de-Broglie wavelengths of the particles $\frac{\lambda_1}{\lambda_2}$ is

(A) $\frac{m_2}{m_1}$

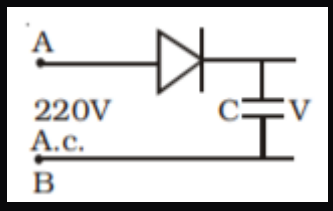
(B) $\frac{m_1}{m_2}$

(C) $\frac{\sqrt{m_1}}{\sqrt{m_2}}$

(D) 1 : 1

Q.13

A 220 V A.C. supply is connected between points A and B. What will be the potential difference V across the capacitor? Assume ideal diode



(A) 220V.

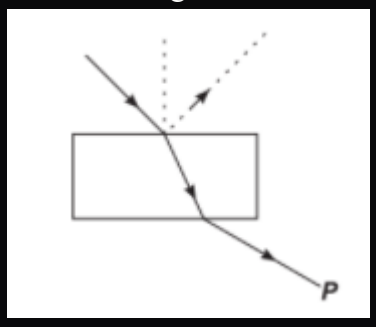
(B) 110V.

(C) 0V.

(D) $220\sqrt{2}\text{ V}$.

Q.14

Consider a light beam incident from air to a glass slab at Brewster's angle as shown in the figure.



A polaroid is placed in the path of the emergent ray at a point P and rotated about an axis passing through the centre and perpendicular to the plane of the polaroid.

(A) For a particular orientation there shall be darkness as observed through the polaroid

(B) The intensity of light as seen through the polaroid shall be independent of the rotation

(C) The intensity of light as seen through the polaroid shall go through a minimum but not zero for two orientations of the polaroid

(D) The intensity of light as seen through the polaroid shall go through a minimum for four orientations of the polaroid

Q.15

The dimensional formula of $\frac{\varepsilon_0 d\phi_E}{dt}$ is :

(A) ML^2T^{-3}

(B) $[A]$

(C) $\left[ML^2A^{-1}T\right]$

(D) $\left[ML^{-1}T^{-2}\right]$

Q.16

Let A and B are two identical objects of same mass initially are charged electrostatically by rubbing against each other. Let A has got a positive charge and B has got a negative charge, then match column I and column II.

	Column I		Column II
(i)	Mass of A	a.	Increases
(ii)	Mass of B	b.	Decreases
(iii)	Work function of A	c.	Is less
(iv)	Work function of B	d.	Is more

(A) (i-d), (ii-c), (iii-b), (iv-a)

(B) (i-b), (ii-d), (iii-c), (iv-a)

(C) (i-b), (ii-a), (iii-c), (iv-d)

(D) (i-c), (ii-d), (iii-a), (iv-b)

Q.17

One mole of diatomic gas is being heated in a closed tank from $300\ \mathrm{K}$ up to $1000\ \mathrm{K}$. During the process part of the molecules dissociate. At $1000\ \mathrm{K}$ the energy of the diatomic molecules are only half of that of the whole gas. By what factor has the gas pressure increased $\left(\mathrm{P}_{\text{final}}/\mathrm{P}_{\text{initial}}\right)$? (The oscillation of the molecules are not to be taken in account).

(A) $\frac{160}{33}$

(B) $\frac{16}{11}$

(C) $\frac{3}{2}$

(D) None

Q.18

A star has a supply of 10^{50} deuterons. It produces energy via a fusion reaction
${}_1^2\mathrm{H} + {}_1^2\mathrm{H} \longrightarrow {}_1^3\mathrm{H} + p$
and ${}_1^2\mathrm{H} + {}_1^3\mathrm{H} \longrightarrow {}_2^4\mathrm{He} + n$
where, masses of nuclei are $m\left({}^2\mathrm{H}\right) = 2.014\mathrm{amu}, m(p) = 1.007\mathrm{amu}$
$m(n) = 1.008\mathrm{amu}$ and $m\left({}^4\mathrm{He}\right) = 4.001\mathrm{amu}$
If the average power radiated by the star is $10^{16}\ \mathrm{W}$. The deuteron supply of the star is exhausted in a time of the order of $(1\mathrm{amu} = 931\mathrm{MeV})$

(A) 10^{22}

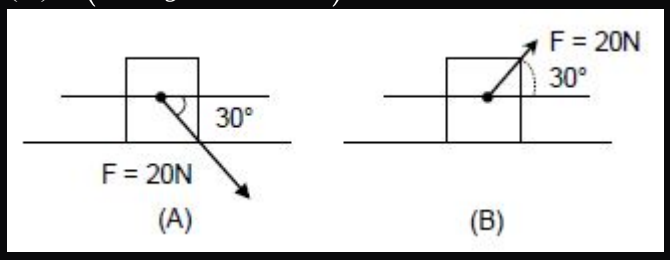
(B) 10^{18}

(C) 10^{8}

(D) 10^{12}

Q.19

Two cases shown in figure. A block of mass 5 kg is (i) pushed in case (A) and (ii) pulled in case (B), by a force $\mathrm{F} = 20\mathrm{N}$, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $\mu = 0.2$. what is the difference between the accelerations of the block, in case (B) and case (A) ? $(take\ g = 10ms^{-1})$



(A) $0.8\,\mathrm{ms}^{-2}$

(B) $0\,\mathrm{ms}^{-2}$

(C) $3.2\,\mathrm{ms}^{-2}$

(D) $0.4\,\mathrm{ms}^{-2}$

Q.20

In a vernier callipers if each centimetre of the main scale is divided into q equal parts and p vernier scale divisions coincide with $(p-1)$ main scale divisions, then what will be the least count?

(A) $\frac{p}{q}(p-1)$

(B) $\frac{q}{p}$

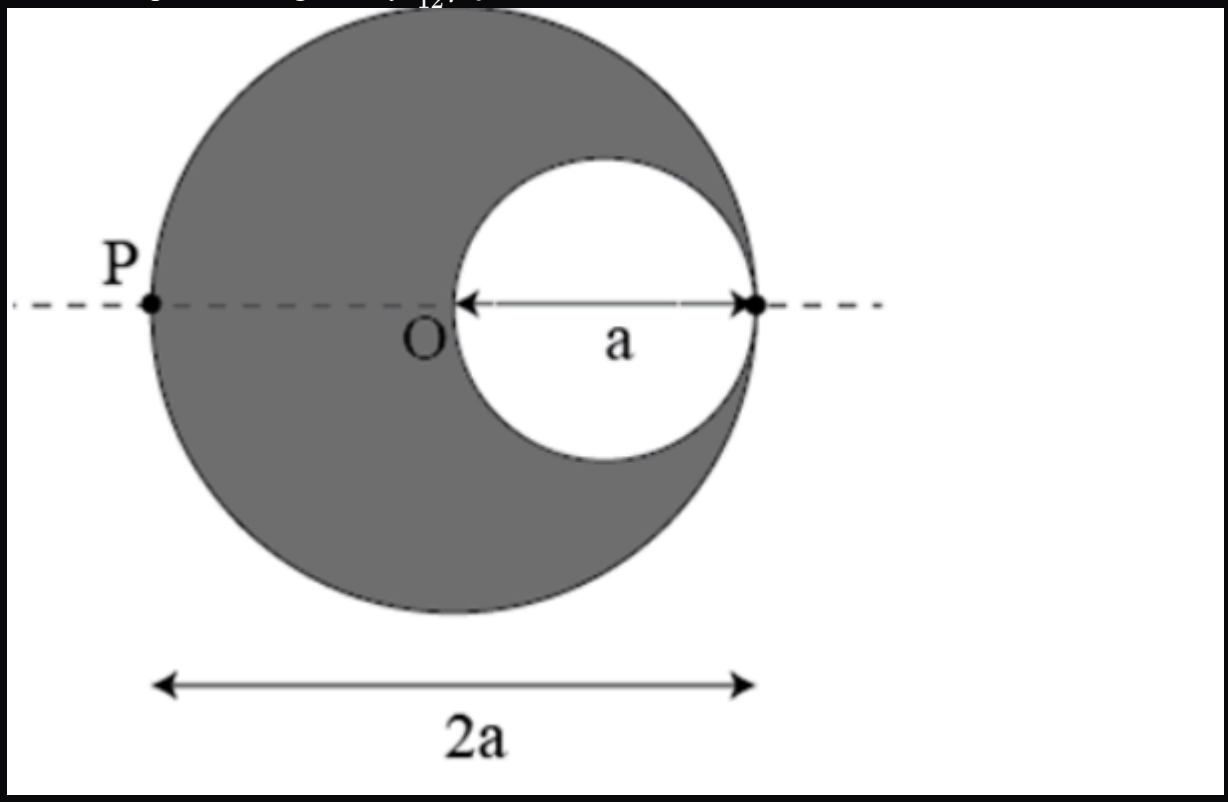
(C) $\frac{q-1}{pq}$

(D) $\frac{1}{pq}$

Q.21

A cylindrical cavity of diameter a exists inside a cylinder of diameter $2a$ as shown in the figure. Both the cylinder and the cavity are infinity long. A uniform current density J flows along the length. If the magnitude of the magnetic

field at the point P is given by $\frac{N}{12}\mu_0 aJ$, then the value of N is



Q.22

If potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{\mathrm{a}}{x^8} - \frac{\mathrm{b}}{x^4}$, where a and b are constants in standard SI units and x is in metres. Find the dissociation energy of the molecule (in J). [Take $\mathrm{a} = 4\ \mathrm{J\ m^8}$ and $\mathrm{b} = 20\ \mathrm{J\ m^4}$]

Q.23

The time taken by a particle performing SHM to pass from point A to B where its velocities are same is $2s$. After another $2s$ it returns to B. The time period of oscillation is: (in seconds)

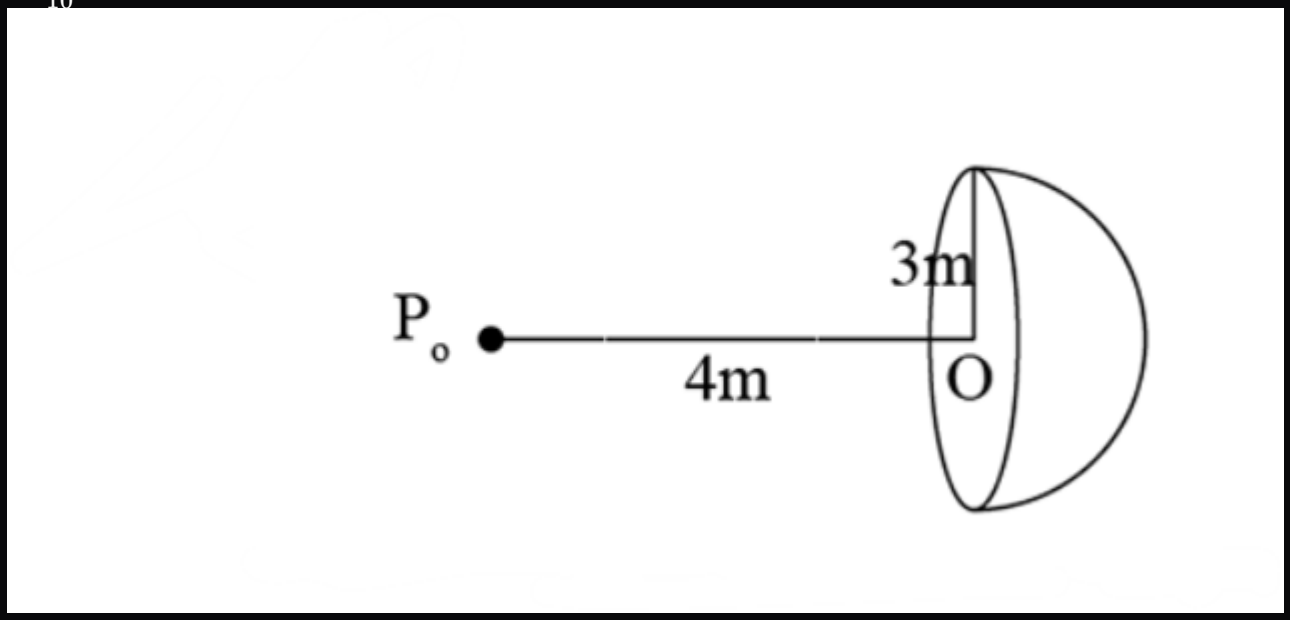
Q.24

The temperature at which a molecule of nitrogen will have the same rms velocity as a molecule of oxygen at $127^\circ\mathrm{C}$ is (Mark answer in Kelvin) (Roundoff answer to nearest integer).

Q.25

A point source of light of power P_0 is placed at a distance of $4\ \mathrm{m}$ from the centre of a thin hemispherical shell as shown in the figure, The shell has a radius of $3\ \mathrm{m}$ and it behaves like a perfect black body. If the temperature of the hemisphere is related to the power of the source as $T^4 = \frac{P_0}{n\pi\sigma}$ where σ is the Stefan's constant, then find the value

of $\frac{n}{10}$.



Q.26

The carbonyl metal complex which cannot exist:

(A) $\mathrm{Cr(CO)_6}$

(B) $\mathrm{Mn(CO)_5}$

(C) $\mathrm{Ni(CO)_4}$

(D) $\mathrm{Fe(CO)_5}$

Q.27

Lucas test is used to make distinction between $1^\circ, 2^\circ$ and 3° alcohols.

$$\mathrm{ROH} + \underset{\text{conc}}{\mathrm{HCl}} \xrightarrow{\text{anhyd } \mathrm{ZnCl_2}} \underset{\substack{\text{white} \\ \text{turbidity}}}{\mathrm{RCl}\downarrow} + \mathrm{H_2O}$$

Which of the following statements is INCORRECT?

(A) ROH behaves as base

(B) Greater the value of $\mathrm{pK_a}$ (alcohol), greater the reactivity with conc. HCl and thus sooner the formation of white turbidity

(C) Alcohol which reacts fastest with Na metal, will give turbidity at fastest rate.

(D) Alcohol which gives red colour during Victor Meyer test will give turbidity at slower rate than those giving blue or white colour during Victor Meyer test.

Q.28

For reactions $\mathrm{A} \rightarrow \mathrm{B}$ and $\mathrm{P} \rightarrow \mathrm{Q}$ Arrhenius constants are 10^8 and 10^{10} respectively. If $\mathrm{E_{A \rightarrow B}} = 600 \text{ cal/mole}$ and $\mathrm{E_{P \rightarrow Q}} = 1200 \text{ cal/mole}$, then find the temperature at which their rate constants are same.

(Given: $\mathrm{R} = 2 \text{ cal/mole/K}$)

(A) 600 K

(B) $300 \times 4.606\ \mathrm{K}$

(C) $\frac{300}{4.606}\mathrm{K}$

(D) $\frac{4.606}{600}\mathrm{K}$

Q.29

Consider the following reactions, which are carried out at the same temperature.

$$\mathrm{CH_3 - Br + OH^{\ominus} \xrightarrow{EtOH} CH_3 - OH + Br^{\ominus}} \quad (\mathrm{i})$$

$$\mathrm{CH_3 - Br + OH^{\ominus} \xrightarrow{DMSO} CH_3 - OH + Br^{\ominus}} \quad (\mathrm{ii})$$

Which of the following statement is correct about these reactions?

(A) Both the reactions take place at the same rate

(B) The second reaction takes place faster than first reaction

(C) The first reaction takes place faster than second reaction

(D) Both the reactions take place by $\mathrm{S_N1}$ mechanism.

Q.30

Which of the following set of compounds represents isostructural characteristics?

(A) $\mathrm{SF_4, CH_4, NH_3}$

(B) $\mathrm{NF_3, BCl_3, NH_3}$

(C) $\mathrm{BF_3, NF_3, AICl_3}$

(D) $\mathrm{BF_3, BCl_3, BBr_3}$

Q.31

Which of the following represents the correct order of increasing first electron gain enthalpy (modulus value) for $\mathrm{O, S, F}$ and Cl ?

(A) $\mathrm{O < S < F < Cl}$

(B) $\mathrm{S < O < F < Cl}$

(C) $\mathrm{O < S < Cl < F}$

(D) $F < S < O < Cl$

Q.32

Choose the correct statement among the following:

(A) Ψ^2 represents the electron probability density.

(B) The number of peaks in radial distribution is $\mathrm{n - l.}$

(C) A node is a point in space around the nucleus where the wave function Ψ is zero.

(D) All of the above.

Q.33

What will be the nature of existence of an amino acid (containing one amino and one carboxylic acid group) in solution of pH < pK_{a_1} ?

(A) It exists an anion

(B) It exists as cation

(C) It exists as zwitter ion

(D) It exists as neutral species with no charge

Q.34

For distinction between CH_3CHO and C_6H_5CHO the reagent used is :

(A) KCN

(B) HCN

(C) NH_2OH

(D) PCl_5

Q.35

The equivalent conductivity of $N/10$ solution of acetic acid at $25^\circ C$ is $14.3 ohm^{-1}\ cm^2 eq^{-1}$. What will be the degree of dissociation of acetic acid?
$\left(\Lambda^\infty_{CH_3COOH} = 390.71 ohm^{-1}\ cm^2 eq^{-1}\right)$

(A) 3.66%

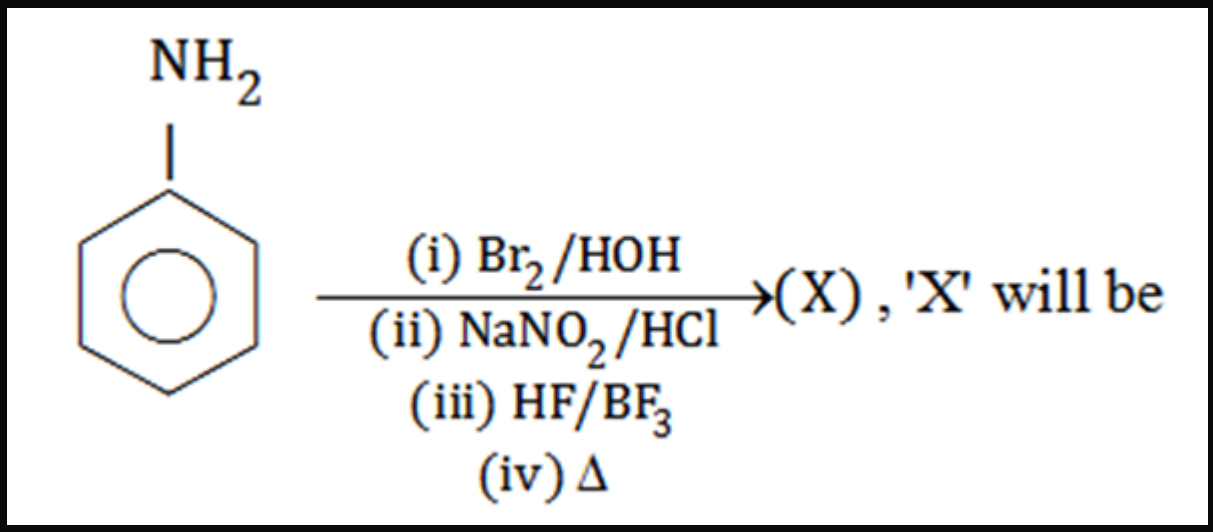
(B) 3.9%

(C) 2.12%

(D) 0.008%

Q.36

In the reaction



(A) p-Bromofluorobenzene

(B) 2, 4, 6-Tribromofluorobenzene

(C) p-Bromoaniline

(D) 1, 3, 5-Tribromobenzene

Q.37

The dissociation constant of two acids HA_1 and HA_2 are 3.14×10^{-4} and 1.96×10^{-5} respectively. The relative strength of the acids will be approximately

(A) $1 : 4$

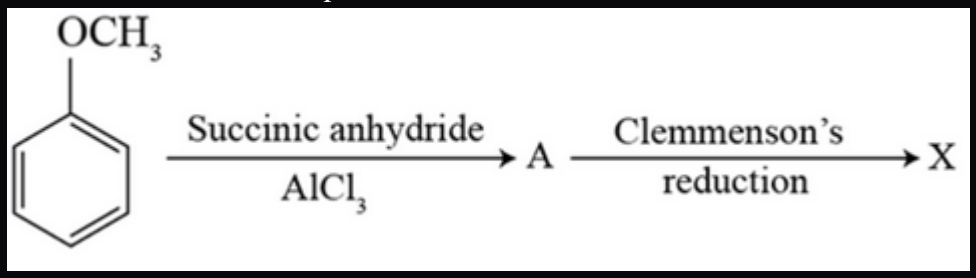
(B) $4 : 1$

(C) $1 : 16$

(D) $16 : 1$

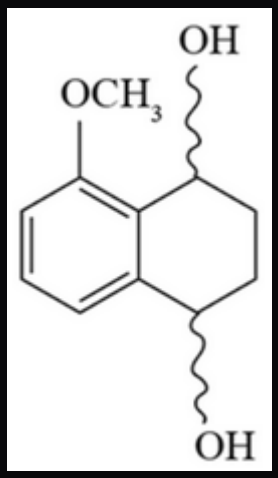
Q.38

Consider the reaction sequence below:

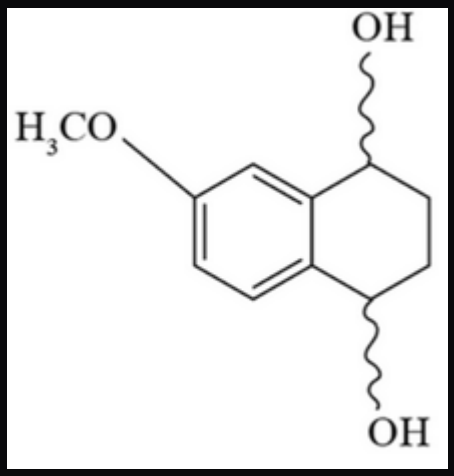


Compund X is :

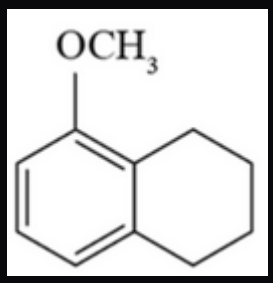
(A)



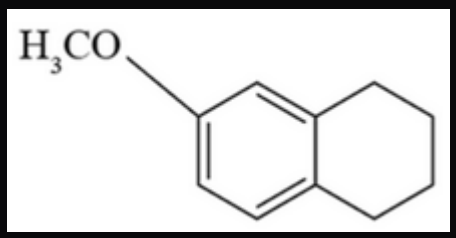
(B)



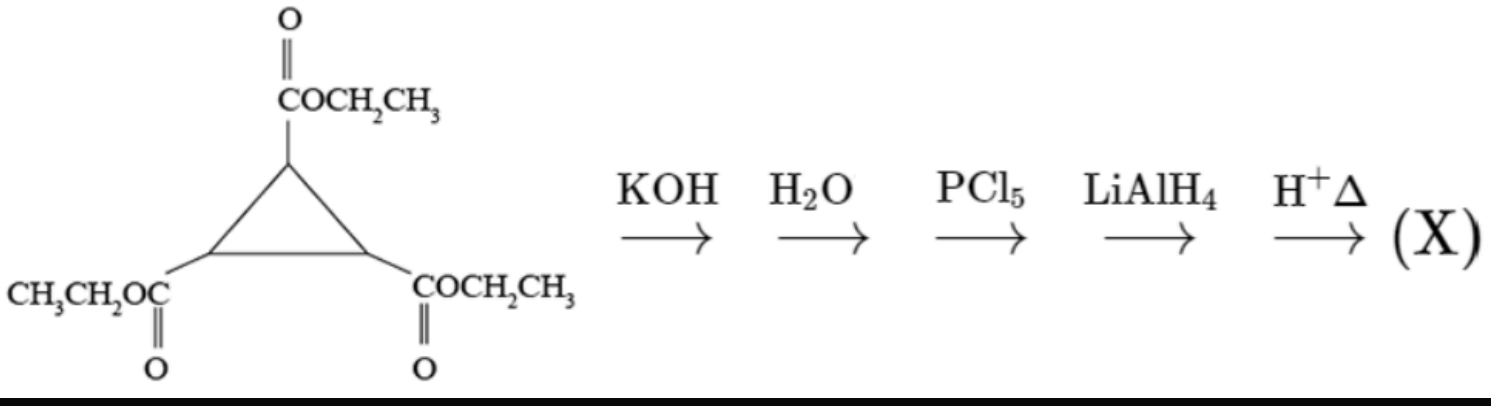
(C)



(D)

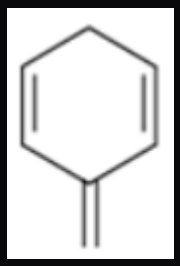


Q.39

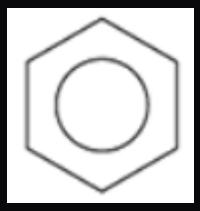


Product (X) is

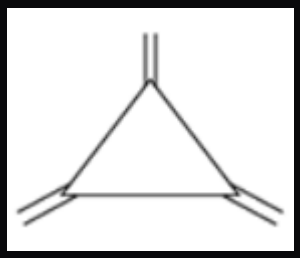
(A)



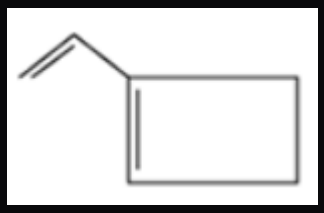
(B)



(C)



(D)



Q.40

A 100ml solution of $0.1\text{ N}-\text{HCl}$ was titrated with $0.2\text{ N}-\text{NaOH}$ solution. The titration was discontinued after adding 30ml of NaOH solution. The remaining titration was completed by adding $0.25N-KOH$ solution. The volume of KOH required for completing the titration is

(A) 16ml

(B) 32ml

(C) 35ml

(D) 70ml

Q.41

Assertion : Given x mL of a sodium bicarbonate ($NaHCO_3$) solution requires y mL of hydrochloric acid (HCI), for complete neutralization. x mL of the same solution would require, for complete neutralization, y mL of sodium hydroxide (NaOH), whose molarity is same as that of hydrochloric acid (HCI).
Reason : sodium bicarbonate($NaHCO_3$) as an amphoteric salt that can be neutralized by either a strong acid or a strong base.

(A) Both assertion and reason are correct, but reason is not the correct explanation of assertion.

(B) Assertion is correct, but reason is wrong.

(C) Assertion is wrong, but reason is correct.

(D) Both assertion and reason are correct and reason is the correct explanation of the assertion.

Q.42

Assertion (A) : $AlCl_3$ exists as a dimer through halogen bridged bonds.
Reason (R) : $AlCl_3$ gets stability by accepting electrons from the bridged halogen

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true, but (R) is not true

(D) (A) is not true, but (R) is true

Q.43

Cane sugar on hydrolysis yields _____.

(A) glucose and maltose

(B) glucose and lactose

(C) glucose and fructose

(D) only glucose

Q.44

Essential Condition for molecules to show ring isomerism:
i. Minimum number of carbon should be 3
ii. Minimum number of carbon should be 4
iii. At least 1 degree of unsaturation must be present.
iv. At least 2 degree of unsaturation must be present.

(A) i and iii

(B) ii and iii

(C) i and iv

(D) ii and iv

Q.45

Lanthanoid which has the smallest size in $+3$ state is

(A) Tb

(B) Er

(C) Ce

(D) Lu

Q.46

$\frac{x}{20}M$ concentration of H^+ion must be maintained in a saturate $H_2S(0.1M)$ to precipitate CdS but not ZnS, if $\left[\mathrm{Cd}^{2+}\right]=\left[\mathrm{Zn}^{2+}\right]=0.1\mathrm{M}$ initially.
$\mathrm{K}_{\mathrm{sp}}(\mathrm{CdS})=8\times10^{-27},\ \mathrm{K}_{\mathrm{sp}}(\mathrm{ZnS})=1\times10^{-21}\ \mathrm{K}_{\mathrm{a}}\left(\mathrm{H}_2\ \mathrm{S}\right)=1\times10^{-21}.\mathrm{ZnS}$
will not precipitate at a concentration of H^+greater than $\frac{\mathrm{x}}{20}\mathrm{M}$. The value of $\mathbf{x}$ is.

Q.47

An organic compound undergoes first order decomposition. The time taken for its decomposition to $\frac{1}{8}$ and $\frac{1}{10}$ of its initial concentration are $\mathrm{t}_{\frac{1}{8}}$ and $\mathrm{t}_{\frac{1}{10}}$ respectively. What is the value of $\frac{\left[\mathrm{t}_{\frac{1}{8}}\right]}{\mathrm{t}_{\frac{1}{10}}}\times10$?

(take $\log_{10}2=0.3$)

Q.48

The value of P° for benzene is $640\ \mathrm{mm}$ of Hg. The vapour pressure of solution containing $2.5\ \mathrm{g}$ substance in $39\ \mathrm{g}$ benzene is $600\ \mathrm{mm}$ of Hg. What is the molecular mass of X ?

Q.49

$$\underset{(\mathrm{A})}{\mathrm{C_{12}H_{16}}}\xrightarrow{i)\mathrm{O_3}\quad ii)\mathrm{Zn,H_2O_2}}2\mathrm{H_3C-\overset{\overset{\mathrm{O}}{\|}}{C}-CH_3}+2\mathrm{HOOC-\overset{\overset{\mathrm{O}}{\|}}{C}-COOH}$$

What is degree of unsaturation of compound A?

Q.50

Consider the following reversible reaction:
$\mathrm{NO}+\mathrm{NO_3}\rightleftharpoons2\mathrm{NO_2}$

If 1.0 mol of NO is mixed with 3.0 mol of NO_3 , 'x' mole of NO_2 is produced at equilibrium. If 2.0 mol of NO is added further, 'x' mol of NO_2 is further produced. What is the value of equilibrium constant?

Q.51

The solution of $ydx - xdy + xy^2dx = 0$, is

(A) $\frac{x}{y} + x^2 = \lambda$

(B) $\frac{x}{y} + \frac{x^2}{2} = \lambda$

(C) $\frac{x}{2y^2} + \frac{x^2}{4} = \lambda$

(D) None of these

Q.52

The value of the definite integral $\int_e^{e^{2010}} \frac{1}{x}\left(1 + \frac{1-\ln x}{\ln x \cdot \ln\left(\frac{x}{\ln x}\right)}\right) dx$, equals:

(A) $2009 - \ln(2010 - \ln 2010)$

(B) $2010 - \ln(2009 - \ln 2009)$

(C) $2009 - \ln(2010 - \ln 2009)$

(D) $2010 - \ln(2010 - \ln 2010)$

Q.53

Suppose $n \geq 3$ person are sitting in a row. Two of them are selected at random. The probability that they are not together is

(A) $1 - \frac{2}{n}$

(B) $\frac{2}{n-1}$

(C) $1 - \frac{1}{n}$

(D) None of these

Q.54

For a positive integer m, if $\lim\limits_{x\to\infty}\left(x^3\ \ln\left(\frac{x+1}{x}\right) + \frac{x}{2} - x^2\right) = \frac{1}{m}$. Then the value of m is

(A) 1

(B) 2

(C) 3

(D) 4

Q.55

Let A and B are two non-singular matrices of order 3 such that $A+B=2I$ and $A^{-1}+B^{-1}=3I$, then AB is equal to (where, I is the identity matrix of order 3)

(A) A

(B) B

(C) $\frac{2I}{3}$

(D) $2I$

Q.56

The $S.D.$ of the following data is nearly

x_i	140	145	150	155	160	165	170	175
f_i	4	6	15	30	36	24	8	2

(A) 8.64

(B) 7.26

(C) 7.05

(D) None

Q.57

The integral $I=\int\limits_0^{100\pi}\left[\tan^{-1}x\right]dx$, (where, $[.]$ represents the greatest integer function) has the value $K\pi+\tan p$, then the value of $K+p$ is equal to

(A) 101

(B) 99

(C) 100π

(D) 99π

Q.58

Let$P(x)=x^6-x^5-x^3-x^2-x\ \ \&\ \ Q(x)=x^4-x^3-x^2-1$. If α,β,γ and δ are the roots of $Q(x)=0$ then the value of $P(\alpha)+P(\beta)+P(\gamma)+P(\delta)$ is equal to

(A) 4

(B) 6

(C) 8

(D) 12

Q.59

If $f''(x)>0,\ \forall x\in R,\ f'(3)=0$ and $g(x)=f\left(\tan^2x-2\tan x+4\right),\ 0<x<\frac{\pi}{2}$, then $g(x)$ is increasing in the interval

(A) $\left(0, \frac{\pi}{4}\right)$

(B) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

(C) $\left(0, \frac{\pi}{3}\right)$

(D) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Q.60

Let $f : A \to B$ be an one to one function
defined as $f(x) = \frac{\sin^{-1} x + \tan^{-1} x}{\cos^{-1} x + \cot^{-1} x}$ then, the number of solutions of the equation
$f\left(x^3 + 14x^2 + 13x - 5\right) = f\left(1 - x^2 + x^3\right)$ is

(A) 0

(B) 1

(C) 2

(D) 3

Q.61

The ends of the base of an isosceles triangle at $(2, 0)$ and $(0, 1)$ and the equation of one side is $x = 2$ then the orthocenter of the triangle is :

(A) $\left(\frac{2}{3}, \frac{3}{5}\right)$

(B) $\left(\frac{5}{4}, 1\right)$

(C) $\left(\frac{3}{4}, 1\right)$

(D) $\left(\frac{4}{3}, \frac{7}{2}\right)$

Q.62

The general values of θ which satisfies the equation $(\cos\theta + i\sin\theta)(\cos 3\theta + i\sin 3\theta)(\cos 5\theta + i\sin 5\theta)$ $\ldots\ldots(\cos(2n-1)\theta + i\sin(2n-1)\theta) = 1$ (where r is integer) is :

(A) $\frac{r\pi}{n^2}$

(B) $\frac{2r\pi}{n^2}$

(C) $\frac{(2r+1)\pi}{n^2}$

(D) None of these

Q.63

The area bounded by $y = f(x), x = 1, x = 4$ and the x-axis is 12 square units where $f(x)$ is a differentiable function such that $f(1) = 3$ and $f(4) = 8$. If $f(x)$ is a bijective function which is always positive, then the area (in square units) bounded by $y = f^{-1}(x), x = 3, x = 8$ and the x-axis is equal to

(A) 29

(B) 32

(C) 12

(D) 17

Q.64

Two circles with centres (2, 3) and (5, 6) cut orthogonally. If radius of both circles are equal, then radius is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Q.65

If $\alpha,\ \beta,\ \gamma$ are the number of elements in sets A, B and C respectively, and sum of number of all subsets of A, B and C is 28, evaluate the maximum number of ordered triplets $(\alpha,\ \beta,\ \gamma)$ that can be formed.

(A) 1

(B) 2

(C) 4

(D) 6

Q.66

Four vectors $\vec{a},\ \vec{b},\ \vec{c}$ and $\vec{x}$ satisfy the relation $\left(\vec{a}\cdot\vec{x}\right)\vec{b}=\vec{c}+\vec{x}$, where $\vec{a}\cdot\vec{b}\neq 1$. The value of $\vec{x}$ in terms of $\vec{a},\ \vec{b}$ and $\vec{c}$ is equal to

(A) $\dfrac{\left(\vec{a}\cdot\vec{c}\right)\vec{b}-\vec{c}\left(\vec{a}\cdot\vec{b}-1\right)}{\left(\vec{a}\cdot\vec{b}-1\right)}$

(B) $\dfrac{\vec{c}}{\vec{a}\cdot\vec{b}-1}$

(C) $\dfrac{2\left(\vec{a}\cdot\vec{c}\right)\vec{b}+\vec{c}}{\vec{a}\cdot\vec{b}-1}$

(D) None of the above

Q.67

Given $f(x) = \log_{10} x$ and $g(x) = e^{\pi i x}$, where $i = \sqrt{-1}$. If $\phi(x) = \begin{vmatrix} f(x)g(x) & (f(x))^{g(x)} & 1 \\ f(x^2)g(x^2) & (f(x^2))^{g(x^2)} & 0 \\ f(x^3)g(x^3) & (f(x^3))^{g(x^3)} & 1 \end{vmatrix}$, then the value of $\phi(10)$ is

(A) 1

(B) 2

(C) 0

(D) i

Q.68

In a shooting competition a man can score $5,\ 4,\ 3,\ 2,\ 1$ or 0 points for each shot. Then the number of different ways in which he can score 10 in seven shots is

(A) 6538

(B) 6648

(C) 6468

(D) 6236

Q.69

Given that $x \in [0, 1]$ and $y \in [0, 1]$ Let A, be the event of selecting a point (x, y) satisfying $y^2 > x$ and B, be the event selecting a point (x, y) satisfying $x^2 > y$, then

(A) $P(A \cap B) = 0$

(B) $A \subset B$

(C) $2P(A) = 3P(B)$

(D) $P(B) < P(A)$

Q.70

A perpendicular is drawn from the point $P(2,\ 4,\ -1)$ to the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$. The equation of the perpendicular from P to the given line is

(A) $\frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$

(B) $\frac{x+2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$

(C) $\frac{x+2}{-6} = \frac{y-4}{3} = \frac{z+1}{2}$

(D) $\frac{x+2}{6} = \frac{y+4}{3} = \frac{z+1}{2}$

Q.71

The function $f(x)$ satisfies the functional equation $3f(x) + 2f\left(\frac{x+59}{x-1}\right) = 10x + 30$ for all real $x \neq 1$. The value of $f(7)$ is:

Q.72

Let matrix $\mathrm{A} = \begin{bmatrix} \mathrm{x} & \mathrm{y} & -\mathrm{z} \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$, where $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{N}$.
If $|(\mathrm{adj}(\mathrm{adj}(\mathrm{adj}(\mathrm{adj}\,\mathrm{A}))))| = 4^8 \cdot 5^{16}$, then the number of such matrices A is equal to

Q.73

Consider the differential equation $\frac{x+\frac{x^3}{3!}+\frac{x^5}{5!}+\cdots}{1+\frac{x^2}{2!}+\frac{x^4}{4!}+\ldots} = \frac{dx-dy}{dx+dy}$. If $y(0) = 1$ and the solution of the differential equation is of the form $2y\,f(x) = me^{2x} + n$. Evaluate $(m+n)\,f(0)$.

Q.74

Consider the number $N = 10!$. If the number of positive divisors of N is λ, then $\frac{\lambda}{30}$ equals

Q.75

The smallest positive integer ' n ' such that
$\frac{1}{\sin 45^\circ \sin 46^\circ} + \frac{1}{\sin 47^0 \sin 48^\circ} + \cdots\cdots\cdots\cdots + \frac{1}{\sin 133^\circ \sin 134^\circ} = \frac{1}{\sin n^0}$ is

Answers & Solutions

Q.1 Answer:

—

Solution:

Stationary waves are produced by two identical waves travelling in opposite direction in same plane.
$\therefore$ for stationary wave
$Z_1 = A\cos(kx - \omega t)$
$Z_2 = A\cos(kx + \omega t)$

Q.2 Answer:

$W_1 > W_2$

Solution:

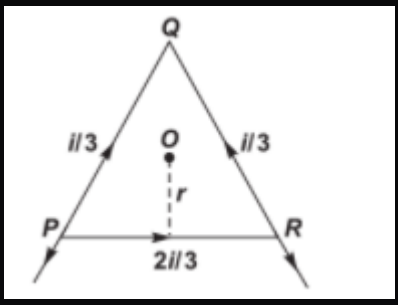
As, work done = area under the $p - V$ diagram
$\therefore \quad W_1 > W_2$

Q.3 Answer:

zero

Solution:

Magnetic field at O due to PR



$$B_1 = \frac{\mu_0}{4\pi}\frac{2i/3}{r}\left[\sin 30^\circ + \sin 30^\circ\right]$$
$$= \frac{\mu_0}{4\pi}\frac{2i}{3r}$$

It is directed outside the paper. Magnetic field at O, due to $PQR(i.e.$, for the wire PQ and $QR)$

$B_2 = 2 \times \frac{\mu_0}{4\pi}\frac{\left(\frac{i}{3}\right)}{\mu}\left[\sin 30^\circ + \sin 30^\circ\right]$

$= \frac{\mu}{4\pi}\frac{2i}{3r}$

It is directed inside the paper.

$\therefore$ Resultant magnetic field at O

$B = B_1 - B_2 = 0$

Q.4 Answer:

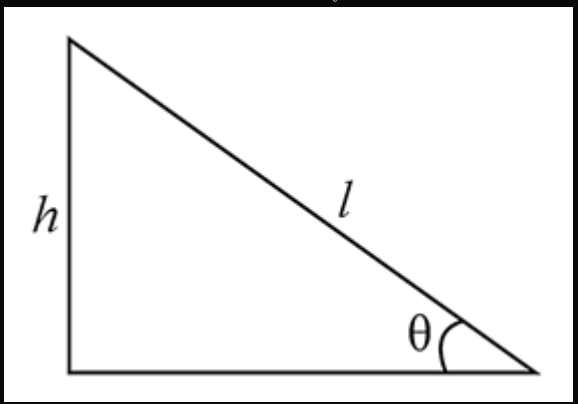
$\tan\left\{\sin^{-1}\left(\frac{h}{l}\right)\right\} = \frac{v^2}{rg}$

Solution:

We know that angle of banking is given by,

$\tan\theta = \frac{v^2}{rg} \quad \ldots(1)$

From triangle, $\sin\theta = \frac{h}{l}$



$\Rightarrow \theta = \sin^{-1}\left(\frac{h}{l}\right) \quad \ldots(2)$

Putting this value in (1), we get, required expression.

$\therefore \quad \tan\left\{\sin^{-1}\left(\frac{h}{l}\right)\right\} = \frac{v^2}{rg}.$

Q.5 Answer:

$\mathrm{T_A} = \mathrm{T_B}$

Solution:

$M_1R_1 = M_2R_2 \quad -(1)$

Force acting on both stars,i.e, gravitational force and centripital forces are balanced.

$\therefore \frac{GM_1M_2}{R^2} = \frac{M_1\,V_1^2}{R_1} = \frac{M_2\,V_2^2}{R_2} - (2)$

Now substitute, $V_1 = \frac{2\pi R_1}{T_1}, \quad V_2 = \frac{2\pi R_2}{T_2}$ **in equation(2) after simplification we get** $\therefore \frac{M_1R_1}{T_1^2} = \frac{M_2R_2}{T_2^2}$

$\Rightarrow T_1 = T_2$

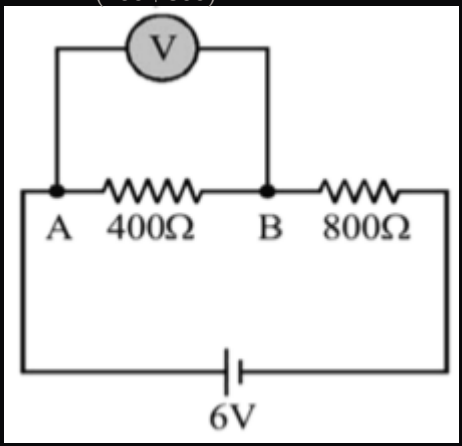
Q.6 Answer:

0.05

Solution:

Before connecting voltmeter potential difference across $400\ \Omega$ **resistance is**

$V_i = \frac{400}{(400+800)} \times 6 = 2V$



After connecting voltmeter equivalent resistance between A **and** $B = \frac{400\times10{,}000}{(400+10{,}000)} = 384.6\Omega$

Hence, potential difference measured by voltmeter

$V_f = \frac{384.6}{(384.6+800)} \times 6 = 1.95\ \text{V}$

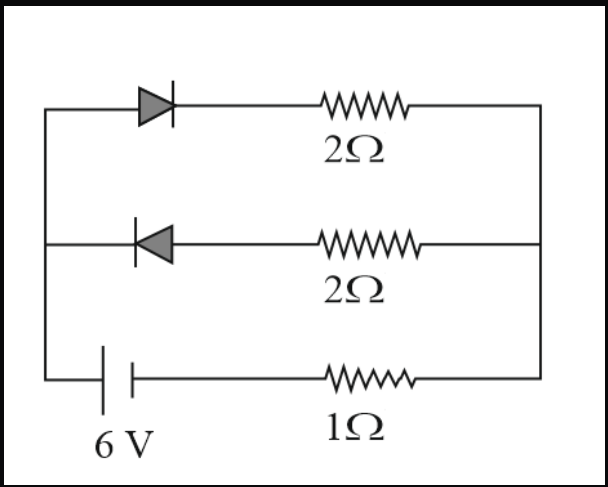
Error in measurement $= V_i - V_f = 2 - 1.95 = 0.05\ \text{V}$

Q.7 Answer:

2

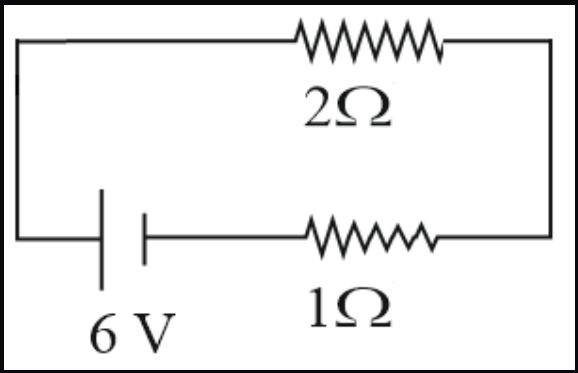
Solution:

We are given a circuit, we need to find the current across $1\ \Omega$ **resistance.**



In the given figure, upper diode is in forward bias and lower diode is in reverse bias.

Thus, the equivalent circuit is,



In forward bias mode, diode behaves like on state and in reverse bias mode, diode behaves like off state.
Current through the circuit , $I = \frac{V}{R}$
$\Rightarrow I = \frac{6}{2+1} = \frac{6}{3} = 2\ \mathrm{A}$

Q.8 Answer:

$-8\mathrm{ms}^{-2}$

Solution:

Given, $v = (180 - 16x)^{1/2}$
squaring both sides $v^2 = 180 - 16x$

Differentiating with respect to t, we get

$$2v\frac{dv}{dt} = 0 - 16\frac{dx}{dt}$$

$$2v\frac{dv}{dt} = -16v$$

$\Rightarrow \frac{dv}{dt} = -8$
$a = -8\ ms^{-2}$
Hence, particle decelerates at the rate of $8\ \mathrm{ms}^{-2}$.

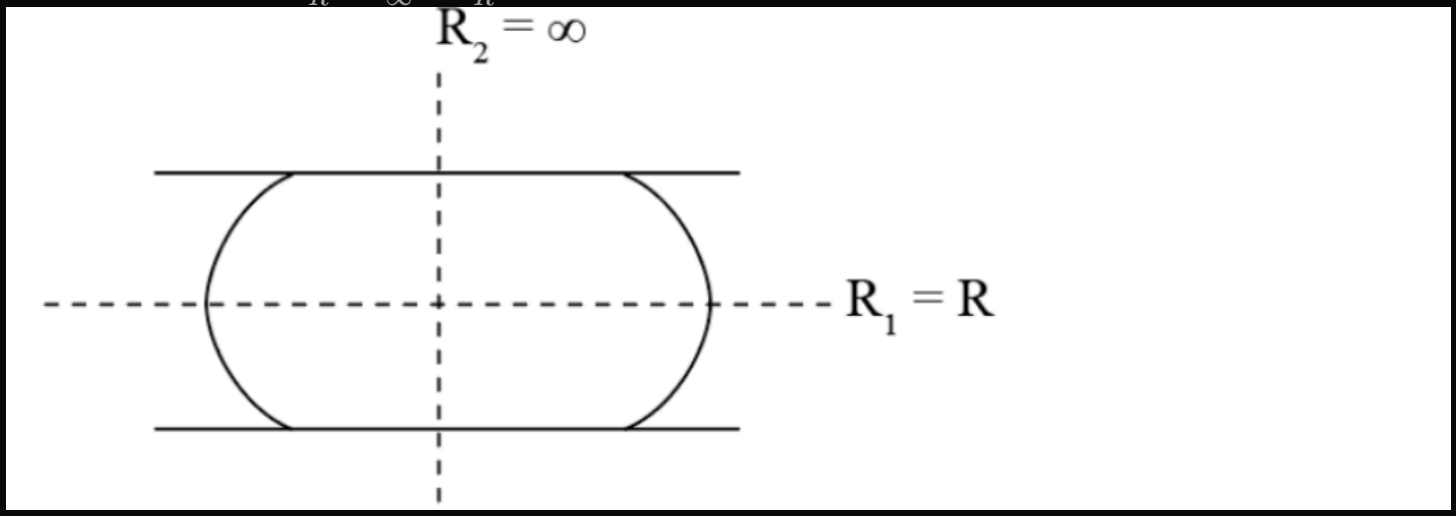
Q.9 Answer:

$\frac{T}{R}$

Solution:

Excess pressure for a liquid drop is given by $P_{\text{inside}} - P_{\text{outside}} = \frac{T}{R_1} + \frac{T}{R_2}$ i.e., Excess pressure $= \frac{T}{R_1} + \frac{T}{R_2}$
where R_1 and R_2 are radii of curvatures of the drop in two mutually perpendicular directions. In this case, $R_1 = R \quad \& R_2 = \infty$

∴ Excess pressure $= \frac{T}{R} + \frac{T}{\infty} = \frac{T}{R}$



Q.10 Answer:

$2.83 \times 10^{-8}\ \mathrm{T}$

Solution:

Given, $E_{\mathrm{rms}} = 6\ \mathrm{V/m}$

$\frac{E_{\mathrm{rms}}}{B_{\mathrm{rms}}} = c \Rightarrow B_{rms} = \frac{E_{\mathrm{rms}}}{c} \ldots (i)$

$\mathrm{B_{rms}} = \frac{\mathrm{B_0}}{\sqrt{2}} \Rightarrow \mathrm{B_0} = \sqrt{2}\,\mathrm{B_{rms}}$

$B_0 = \sqrt{2} \times \frac{E_{\mathrm{rm}}}{C}$ **From equation (i)**

$= \frac{\sqrt{2} \times 6}{3 \times 10^8} = 2.83 \times 10^{-8}\ \mathrm{T}$

Q.11 Answer:

$5\pi V$

Solution:

The emf induced in the second coil is

$\varepsilon = M\frac{di}{dt}$

Where M **is the mutual inductance of two coils,** $\frac{di}{dt}$ **is the rate of change of current in the first coil.**

But $\frac{di = \frac{d}{dt}(i_m \sin \omega t)}{dt} = i_m \omega \cos \omega t \quad \therefore \omega = M i_m \omega \cos \omega t$

For maximum value of emf induced, $\cos \omega t = 1$

∴ The maximum value of the emf induced is $\varepsilon_{\max} = M i_m \omega$

Here, $M = 0.005H, i_m = 10A, \omega = 100\pi \mathrm{rads}^{-1}$

$\varepsilon_{\max} = (0.005H)(10A)(100\pi) = 5\pi V$

Q.12 Answer:

1 : 1

Solution:

By the law of conservation of linear momentum

$Mv = m_1 v_1 + m_2 v_2 \ldots (1)$

As M **is initially at rest** $v = 0$

Now (1) **becomes,** $0 = m_1 v_1 + m_2 v_2$

or $m_1v_1 = -m_2v_2$, it means that the magnitude of the momentum of both masses are equal i.e., $p_1 = p_2$
From de-Broglie hypothesis, $\lambda_1 = \frac{h}{p_1}$ and $\lambda_2 = \frac{h}{p_2}$
So, $\frac{\lambda_1}{\lambda_2} = \frac{p_2}{p_1} = 1$ (as $p_1 = p_2$).

Q.13 Answer:

220 $\sqrt{2}$ V.

Solution:

Potential difference across capacitor will be peak voltage when diode is in forward bias. Diode will be in forward bias when end A is at positive potential of cycle. So potential at
$C = \text{peak value of } V = V_{rms}\sqrt{2} = 220\sqrt{2}\,\text{V}$

Q.14 Answer:

The intensity of light as seen through the polaroid shall go through a minimum but not zero for two orientations of the polaroid

Solution:

According to Brewster's law, the light reflected from the top of glass slab gets polarised. The light refracted into the glass slab and the light emerging from the glass slab is only partially polarised. Therefore, when a polaroid is held in the path of emergent light at P, and rotated about an axis passing through the centre and perpendicular to plane of polaroid, the intensity of light shall go through a minimum but not zero for two orientations of the polaroid.

Q.15 Answer:

$[A]$

Solution:

We have displacement current $I_d = \varepsilon_0 \frac{d\phi_E}{dt}$
So, dimensions of $\varepsilon_0 \frac{d\phi_E}{dt}$ are same as that of current.

Q.16 Answer:

(i-b), (ii-a), (iii-c), (iv-d)

Solution:

As B is negatively charged, due to gain of electrons the mass of B will increase, whereas A is positively charged due to loss of electrons the mass of A will decrease. $m_B > m_A$ Work Function is the minimum energy required by an e- to come onto its surface. As A readily loses its e-, its work function is lesser than B.

Q.17 Answer:

$\frac{160}{33}$

Solution:

Let x mole of the gas dissociate at $1000\ \mathrm{K}$

Number of mole of diatomic gas molecule $= 1 - \mathrm{x}$

Number of moles of monatomic gas molecules $= 2 \times \mathrm{x}$

Energy of diatomic molecules = energy of monatomic molecule

$\Rightarrow (1-x)\frac{5}{2}RT = 2x \times \frac{3}{2}RT$

$\Rightarrow x = 5/11$

Now new number of moles $= (1-x) + 2x = 1 + x = (16/11)$

$P = \frac{nRT}{V}$

Pressure initially at $300\ \mathrm{K} = \mathrm{P_i} = \frac{300\ \mathrm{R}}{\mathrm{V}}$

Pressure finally at $1000\ \mathrm{K} = \mathrm{P_f} = \frac{(1+\mathrm{x})\mathrm{R}\times 1000}{\mathrm{V}} = \frac{16}{11} \times 1000\left(\frac{\mathrm{R}}{\mathrm{V}}\right)$

Ratio $= \frac{160}{33}$

Q.18 Answer:

10^{22}

Solution:

Net process is

$3^2\mathrm{H}_1 \longrightarrow {}^4\mathrm{He}_2 + p + n$

Mass defect, $\Delta m = (3 \times 2.014)$

$-1.007 - 1.008 - 4.001$

$= 0.026\mathrm{amu}$

Energy released by fusion of

3 deuterons

$= 0.026 \times 931\mathrm{MeV}$

$= 24.2\mathrm{MeV}$

$= 3.87 \times 10^{-12}\ \mathrm{J}$

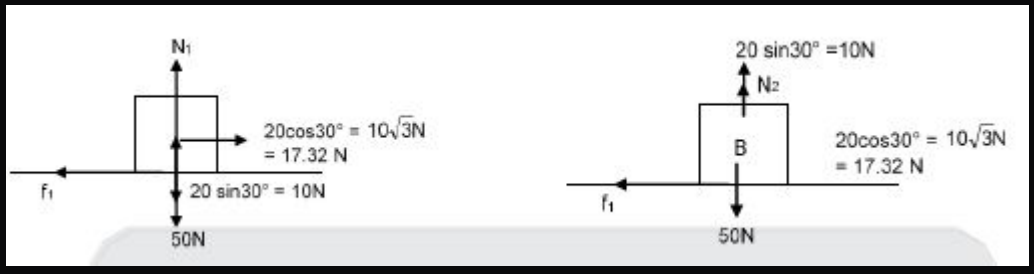
Total energy that can be released by the star is $= \frac{10^{50}}{3} \times 3.87 \times 10^{-12}\ \mathrm{J} = 10^{38}\ \mathrm{J}$

$\therefore$ Time for which it can produce energy at the constant rate is $= \frac{10^{38}}{10^{16}} = 10^{22}\ \mathrm{s}$

Q.19 Answer:

$0.8\,\mathrm{ms}^{-2}$

Solution:



$N_1 = 50 + 10 = 60\mathrm{N}$

$F_{4m} = 0.2 \times 60 = 12\mathrm{N}$

$F_{\text{appred}} > 12N$

$\therefore f_1 = 12N$

$\therefore a_1 = \frac{17.32-12}{5}$

$= \frac{5.32}{5}$

$\therefore \quad a_1 - a_2 = -\frac{4}{5}$

$|a_1 - a_2| = \frac{4}{5} = 0.8 \mathrm{m/s^2}$

$\mathrm{N}_2 = 50 - 10 = 40 \mathrm{N}$

$F_{4m} = 0.2 \times 40 = 8N$

$F_{\text{appled}} > 8N$

$a_2 = \frac{17.32-8}{5}$

Q.20 Answer:

$\frac{1}{pq}$

Solution:

$$1\mathrm{MSD} = \frac{1}{q}$$

$$p \times \mathrm{VSD} = (p-1) \times \mathrm{MSD}$$

$$\Rightarrow 1\mathrm{VSD} = \frac{(p-1)}{pq}$$

$$\mathrm{L.C.} = 1\mathrm{MSD} - 1\mathrm{VSD}$$

$$\Rightarrow \frac{1}{q} - \frac{(p-1)}{pq} = \frac{1}{pq}$$

Q.21 Answer:

—

Solution:

$$\mathrm{B} = \frac{\mu_0\left(\mathrm{J}\pi_{\mathrm{a}^2}\right)}{2\pi_{\mathrm{a}}} - \frac{\mu_0\left(\mathrm{J_a}^2/4\right)}{2\pi\left(\frac{3\mathrm{a}}{2}\right)}$$

$$\mathrm{B} = \frac{5\mu_0 \mathrm{Ja}}{12} = \frac{\mu_0 \mathrm{NJa}}{12}$$

$$\text{So } \mathrm{N} = 5$$

Q.22 Answer:

—

Solution:

$U(x) = \frac{4}{x^8} - \frac{20}{x^4}$

$\therefore \ \frac{dU}{dx} = 0 \ \Rightarrow \ \frac{-8\times4}{x^9} + \frac{20\times4}{x^5}$

$\Rightarrow \ \frac{4}{x^5}\left(20 - \frac{8}{x^4}\right) = 0$

$x = \infty$ **and** $x^4 = \frac{8}{20}$

Also $\frac{d^2U}{dx^2} = +\text{ve} \ \text{ at } \ x^4 = \frac{8}{20}$

$\therefore \ U_{\min} = \frac{4}{x^4}\left(\frac{1}{x^4} - 5\right) = \frac{4\times20}{8}\left(\frac{20}{8} - 5\right)$

$\Rightarrow \ U_{\min} = -\frac{4\times20}{8} \times \frac{20}{8} = -25 \ \mathrm{J}$

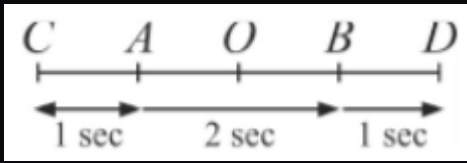
$\therefore$ **Dissociation energy** $= (U_{\text{infinite}} - U_{\min}) = 25 \ \mathrm{J}$

Q.23 Answer:

—

Solution:

$T = 2(1 + 2 + 1)s.$



$= 8$ **second**

Q.24 Answer:

—

Solution:

Root mean square velocity

$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

where R is gas constant, T the temperature and M molecular weight

$\text{Given, } M_{\text{N2}} = 28, \quad M_{\text{O}_2} = 32, \quad T_{\text{O}_2} = 127^\circ\text{C} = 127 + 273 = 400 \text{ K}$

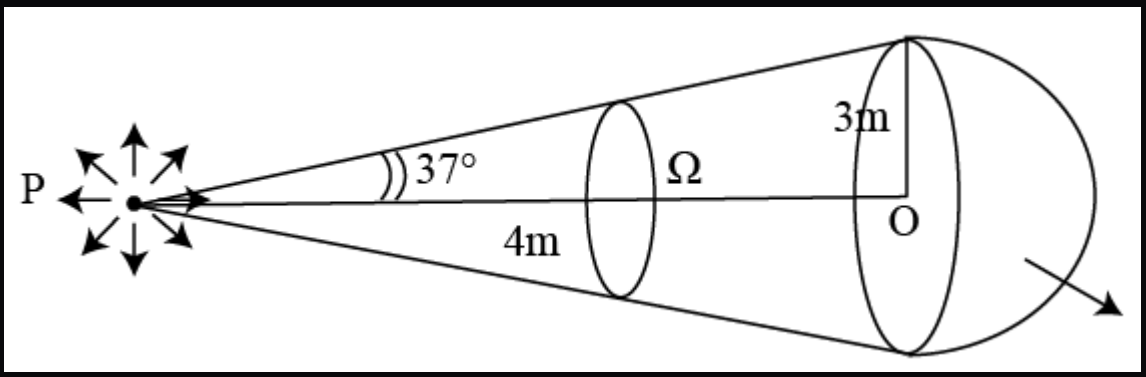
$$\therefore \frac{v_{02}}{v_{\text{N2}}} = \sqrt{\frac{T_{\text{O}_2}}{M_{\text{O}_2}} \times \frac{M_{\text{N}_2}}{T_{\text{N2}}}} = \sqrt{\frac{400}{32} \times \frac{28}{T_{\text{N2}}}} = 1$$

$$\Rightarrow T_{\text{N2}} = 350 \text{ K} = 77^\circ\text{C}$$

Q.25 Answer:

—

Solution:



$$\Omega = 2\pi\left(1-\cos 37^\circ\right) = 2\pi\left(1-\frac{4}{5}\right) = \frac{2\pi}{5}$$

$$P_{\text{received}} = \frac{P_0}{4\pi} \times \frac{2\pi}{5} = \frac{P_0}{10}$$

$$P_{\text{radiated}} = \sigma(1)\left(4\pi R^2\right) T^4 = \frac{P_0}{10}$$

$$T^4 = \frac{P_0}{10 \times 36\pi\sigma} = \frac{P_0}{360\pi\sigma}$$

$$n = 360$$

$$\frac{n}{10} = 36$$

Q.26 Answer:

$\mathrm{Mn(CO)_5}$

Solution:

By applying the EAN rule. The stability of metal carbonyls can be predicted. All the given metal carbonyls except Mn$(CO)_5$ follows the EAN rule and thus, it exists as dimer $\mathrm{Mn_2(CO)_{10}}$.

Q.27 Answer:

Alcohol which reacts fastest with Na metal, will give turbidity at fastest rate.

Solution:

Out of $1^\circ/2^\circ/3^\circ$ alcohols, 1° alcohols react fastest with Na metal which gives turbidity at slowest rate in Lucas test.

Q.28 Answer:

$\frac{300}{4.606}\mathrm{K}$

Solution:

$$\mathrm{A_1 e^{-Ea_1/RT} = A_2 e^{-Ea_2/RT}}$$

$$\mathrm{\frac{A_2}{A_1} = e^{(Ea_2 - Ea_1)/RT}}$$

$$\mathrm{10^2 = Exp\left\{\frac{600}{RT}\right\}\ R = 2\ cal/K-mol}$$

$$\mathrm{2ln10 = \frac{600}{2T}}$$

$$\mathrm{T = \left\{\frac{300}{2\times 2.303}\right\} K}$$

Q.29 Answer:

The second reaction takes place faster than first reaction

Solution:

The preparation of alcohols from alkyl halides is a nucleophilic substitution reaction. EtOH is a polar solvent due to which the nucleophile generated is solvated and the rate of reaction is decreased due to the unavailability of nucleophile for attack.
While, in non-polar solvent such as DMSO, solvation does not occur and all the nucleophiles are available for attack. Hence, the rate of reaction is higher.
Therefore, the second reaction takes place at a faster rate than the first reaction.

Q.30 Answer:

$\mathrm{BF_3}, \mathrm{BCl_3}, \mathrm{BBr_3}$

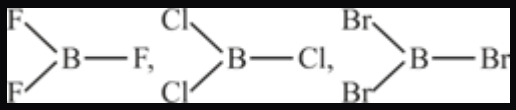
Solution:

As, trihalides of boron have same structure due to same value of H (hybrid orbitals)

$H = \frac{V+M-C+A}{2}$

$= \frac{3+3}{2} = 3$

Hybridisation $= sp^2$



Q.31 Answer:

$\mathrm{O} < \mathrm{S} < \mathrm{F} < \mathrm{Cl}$

Solution:

Modulus value of electron gain enthalpy increases across a period and decreased down the group with an exception that modulus value of electron gain enthalpy of $\mathrm{O} < \mathrm{S}$ and $\mathrm{F} < \mathrm{Cl}$.

Q.32 Answer:

All of the above.

Solution:

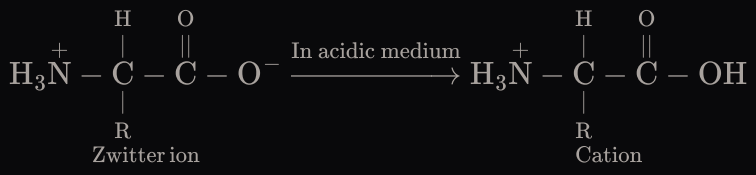
Ψ^2 is a probability factor and represents an atomic orbital.
For $\mathrm{H_2}$ wave function, the number of nodes (the space where probability of finding electron is zero) can be calculated as:
Radial nodes $= (\mathrm{n} - \mathrm{l} - 1)$
The number of peaks is one more than the number of radial nodes, i.e., $\mathrm{n} - \mathrm{l}$.

Q.33 Answer:

It exists as cation

Solution:

At low pH (acidic solution) COO^- ion acts as the base and accepts proton to form the cation.

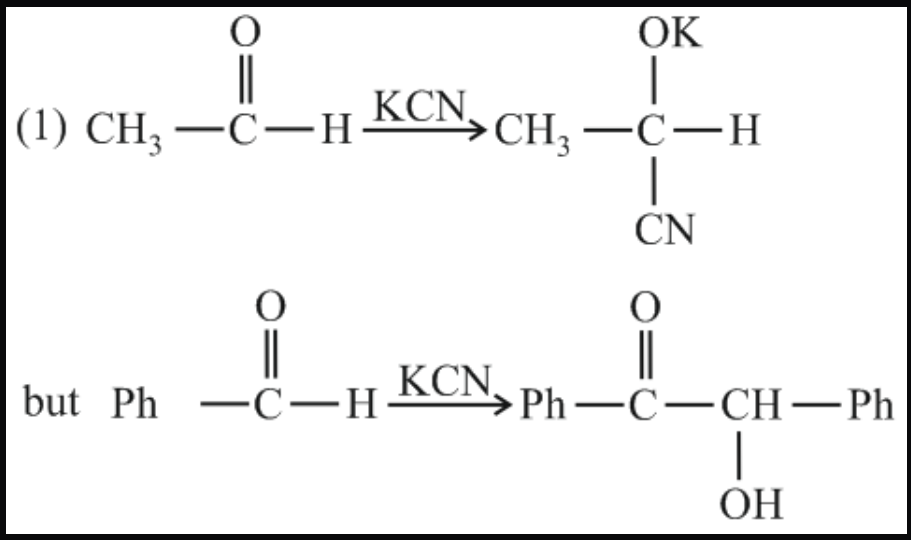


Q.34 Answer:

KCN

Solution:

Aliphatic aldehyde and aromatic aldehyde can be distinguished with the help of KCN.
KCN shows a simple nucleophilic addition initially in both aliphatic and aromatic aldehyde.
But aromatic aldehydes after nucleophilic attack with Cyanide undergoes chargereversal and start to Show electron rich centre charcter at carbonylcarbon which is stabilized by extended Pi conjugation.
Due to nucleophilic character it attacks new electron deficient carbonyl C leads to formation ofa new C-C bond between two aldehyde..and resuts in formation of benzoin.
Thus aromatic aldehyde shows benzoin condesation with KCN and aliphatic aldehydes stop with Cyanohydrins



In other reactions the products are similar.

Q.35 Answer:

3.66%

Solution:

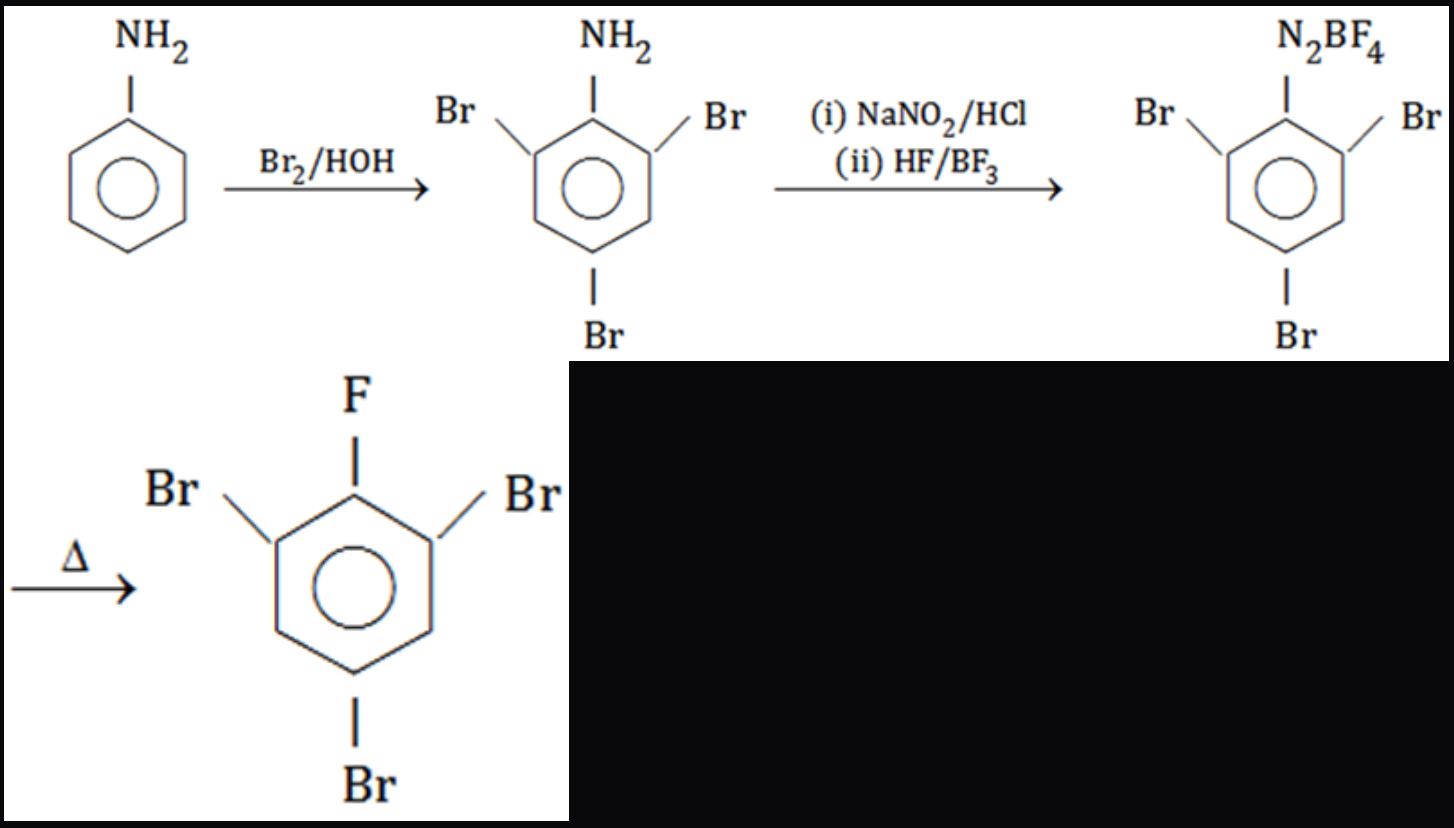
$\Lambda^{\infty}_{CH_3COOH} = 390.71 \text{ohm}^{-1}\ \text{cm}^2\text{eq}^{-1}$
$\Lambda^{\infty}_{CH_3COOH} = 14.3 \text{ohm}^{-1}\ \text{cm}^2\text{eq}^{-1}$
Degree of dissociation $(\alpha) = \frac{\Lambda^c}{\Lambda^{\infty}}$
$= \frac{14.3}{390.71} = 0.0366$ **i.e.** 3.66%

Q.36 Answer:

2, 4, 6-Tribromofluorobenzene

Solution:

Here X is 2, 4, 6-tribromofluorobenzene



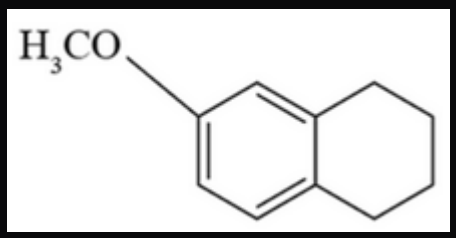
Q.37 Answer:

4 : 1

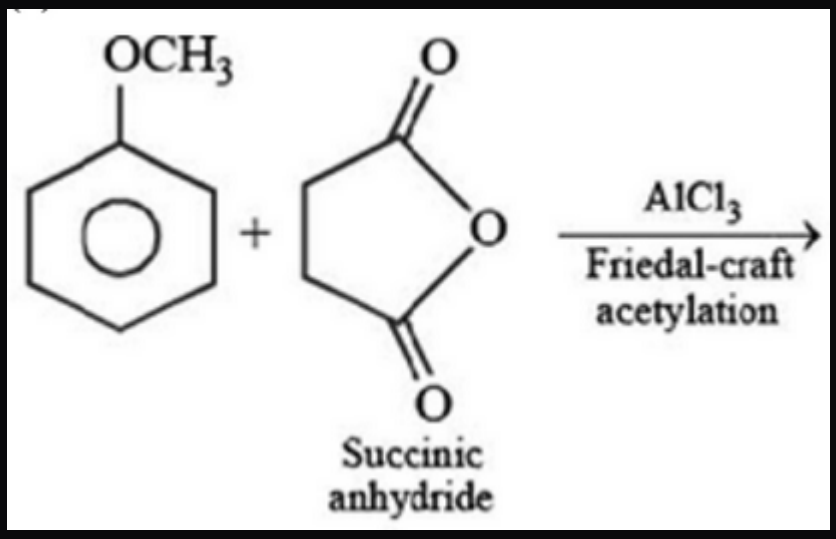
Solution:

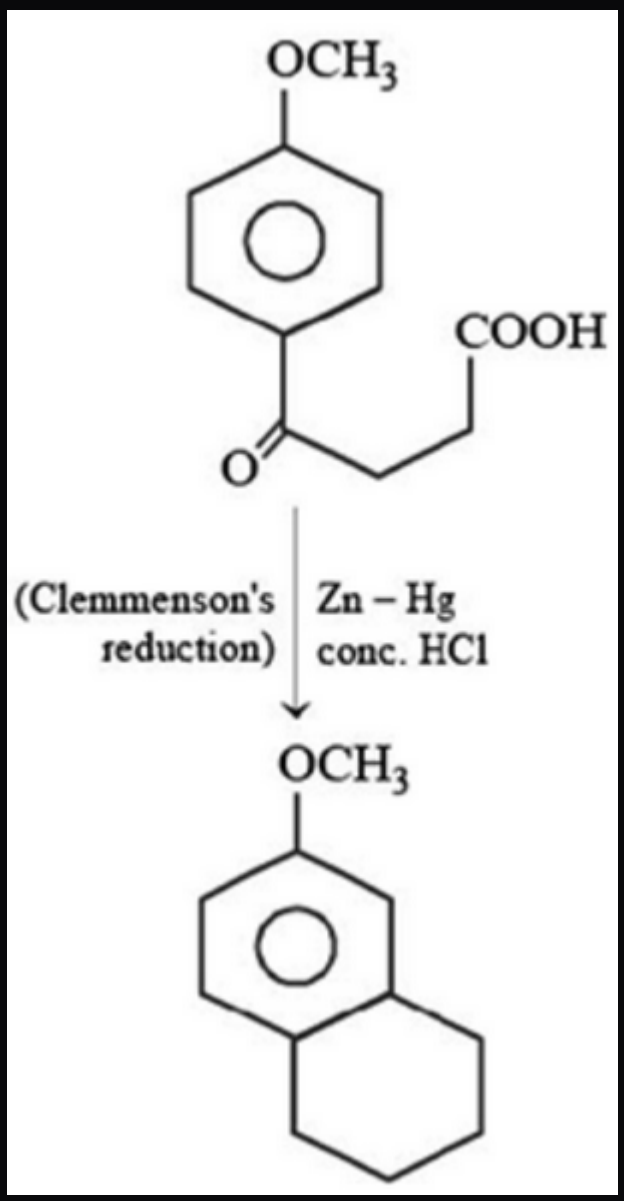
$$\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_{a_1}}{K_{a_2}}} = \sqrt{\frac{3.14\times10^{-4}}{1.96\times10^{-5}}} = 4:1$$

Q.38 Answer:

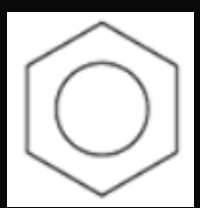


Solution:



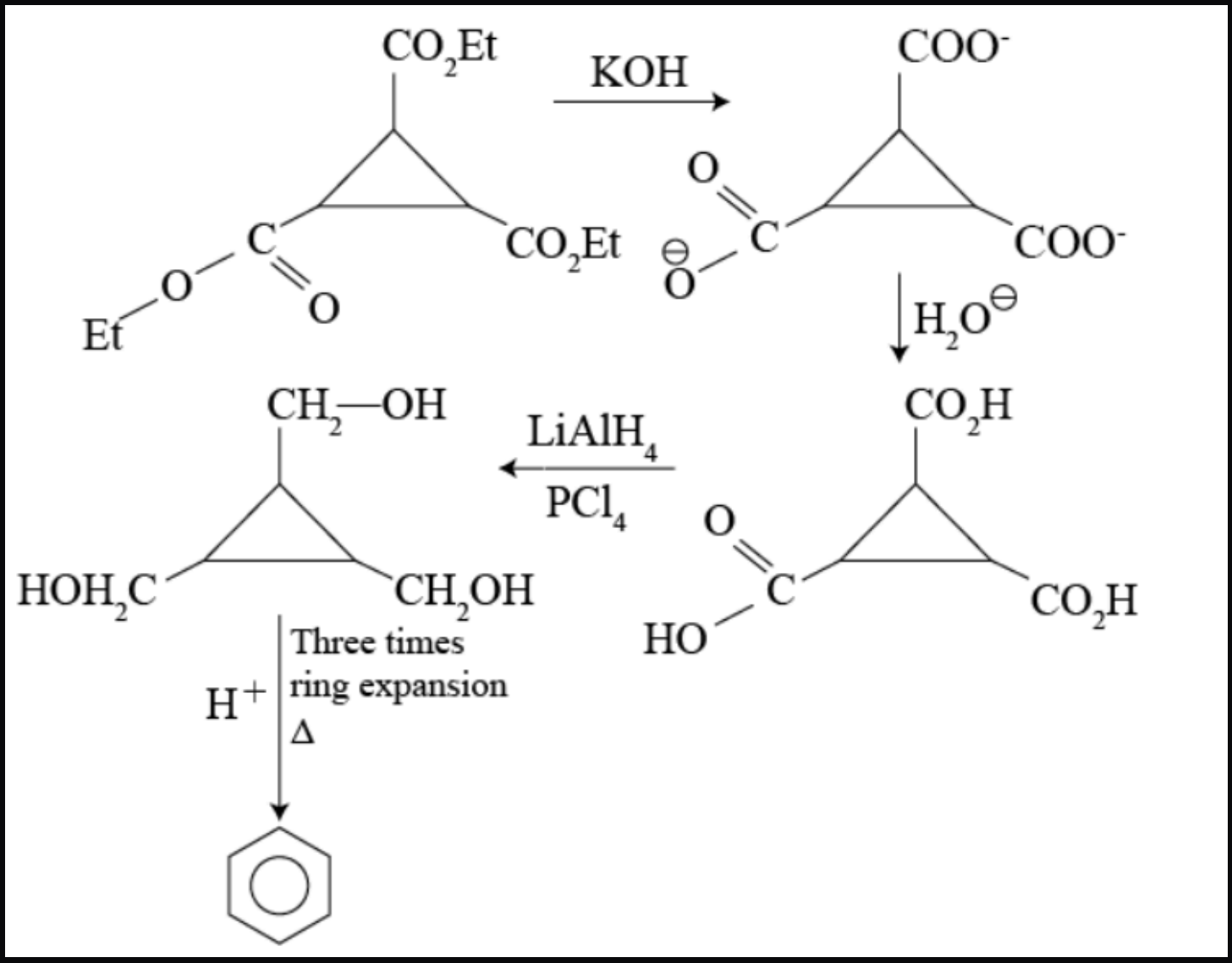


Q.39 Answer:



Solution:

Here Benzene is formed as follows



Q.40 Answer:

16ml

Solution:

In the neutralization of acid and base $N \times V$ of both must be equivalent

$N \times V$ of $HCl = 0.1 \times 100 = 10$

$N \times V$ of $NaOH = 0.2 \times 30 = 6$

$\mathrm{N_1\ V_1(ml) = N \times V(ml) + N \times V(ml)}$

$0.1 \times 100 = 0.2 \times 30 + 0.25 \times V$

$10 = 6 + 0.25V$

$V = \frac{400}{25} \quad V = 16\mathrm{ml}$

Q.41 Answer:

Both assertion and reason are correct and reason is the correct explanation of the assertion.

Solution:

x mL $NaHCO_3$ <≡> y mL of HCl (N) for end point

x mL $NaHCO_3$ <≡> y mL of NaOH (N) for end point

$NaHCO_3 + HCl \longrightarrow NaCl + H_2CO_3$

$NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$

because $NaHCO_3$ an amphoteric salt it can react with strong acid or strong base.

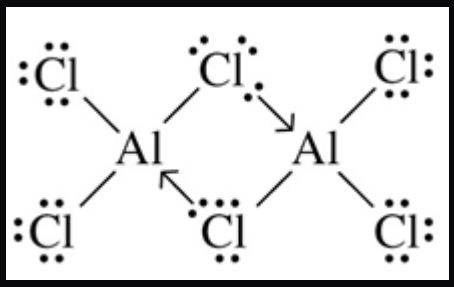
Both statements are correct.

Q.42 Answer:

Both (A) and (R) are true and (R) is the correct explanation of (A)

Solution:

$AlCl_3$ can form dimer and exists as Al_2Cl_6, aluminium has vacant p-orbital which can accommodate electron from chlorine atom. $AlCl_3$ is an electron deficient compound in Al (octet incomplete) thus behaves as Lewis acid and Al completes it by taking electron pair from Cl-atom as shown in figure.



Q.43 Answer:

glucose and fructose

Solution:

Cane sugar is a disaccharide and made up of glucose and fructose. Hence, on hydrolysis it yields these monomers. Cane sugar is also known as sucrose.

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

Q.44 Answer:

i and iii

Solution:

Minimum 3 $C-$ atoms are required to form a ring and atleast one double bond is required to exhibit ring isomerism.

Q.45 Answer:

Lu

Solution:

On going from left to right in lanthanoid series ionic, size decreases i.e.

$Ce^{3+} > Tb^{3+} > Er^{3+} > Lu^{3+}$.

Q.46 Answer:

—

Solution:

$\mathrm{CdS} \rightleftharpoons \mathrm{Cd}_{0.1}^{2+} + \mathrm{S}_{\mathrm{s}}^{2-}$

For precipitation of $\mathrm{CdS}\ \left[\mathrm{Cd}^{2+}\right]\left[\mathrm{S}^{2-}\right] = \mathrm{K}_{\mathrm{sp(CdS)}}$

$0.1 \times \left[\mathrm{S}^{2-}\right] = 8 \times 10^{-27}$

$\left[\mathrm{S}^{2-}\right] = 8 \times 10^{-26}$

For precipitation of $\mathrm{ZnS}\ \left[\mathrm{Zn}^{2+}\right]\left[\mathrm{S}^{2-}\right] = \mathrm{K}_{\mathrm{sp(ZnS)}}$

$0.1 \times \left[\mathrm{S}^{2-}\right] = 10^{-21}$

$\left[\mathrm{S}^{2-}\right] = 10^{-20}$

CdS will start precipitate as the conc of $\left[\mathrm{S}^{2-}\right] = 8 \times 10^{-26}$ **and** ZnS **will start precipitate as the conc of** $\left[\mathrm{S}^{2-}\right] = 10^{-20}$ **so we can raise the concentration of** $\left[\mathrm{S}^{2-}\right]$ **upto** 10^{-20} **Now** $\mathrm{H_2\ S} \rightleftharpoons 2\mathrm{H}^{+} + \mathrm{S}^{2-}$

$\frac{\left(\frac{x}{20}\right)^2 \times 10^{-20}}{0.1} = 10^{-21}$

$\left(\frac{x}{20}\right)^2 = 10^{-2}$

$\frac{x}{20} = 10^{-1}$

$x = 2$

Q.47 Answer:

—

Solution:

As we know

$\mathrm{kt} = \ln \frac{\mathrm{A}_0}{\mathrm{A}_t}$

$\frac{\mathrm{t}_{\frac{1}{8}}}{\mathrm{t}_{\frac{1}{10}}} = \frac{\ln 8}{\ln 10} = 0.9$

$\frac{\mathrm{t}_{\frac{1}{8}}}{\mathrm{t}_{\frac{1}{10}}} \times 10 = 9$

Q.48 Answer:

—

Solution:

$\frac{p^0 - p_s}{p^0} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$

$\frac{640 - 600}{640} = \frac{2.5/M}{\frac{2.5}{M} + \frac{39}{78}}$

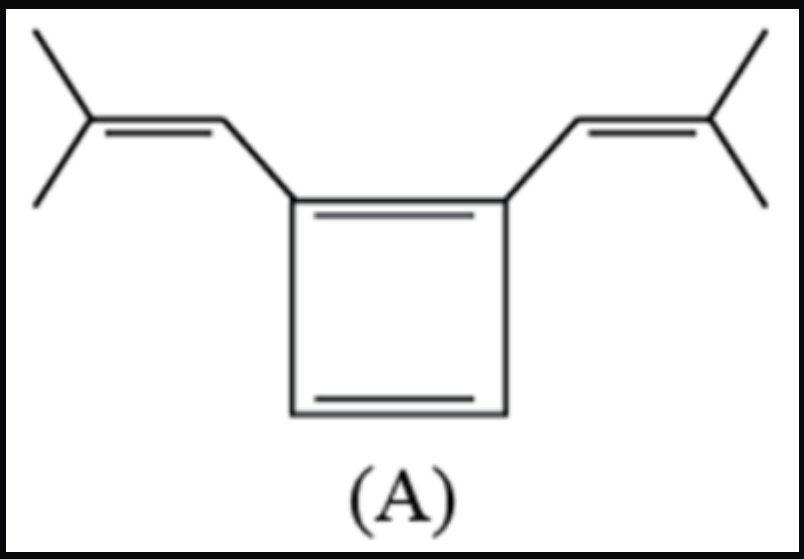
$M = 75$

Q.49 Answer:

—

Solution:

Here degree of unsaturation is 5 (4 double bond and 1 cyclic ring)



(A)

Q.50 Answer:

—

Solution:

$$\begin{array}{ccccccc} \mathrm{NO} & + & \mathrm{NO_3} & \rightleftharpoons & 2\mathrm{NO_2} & ; & \mathrm{K_c} \\ 1-\mathrm{x}/2 & & 3-\mathrm{x}/2 & & \mathrm{x} & & \end{array}$$

$$\begin{array}{ccccccc} \mathrm{NO} & + & \mathrm{NO_3} & \rightleftharpoons & 2\mathrm{NO_2} & ; & \mathrm{K_c} \\ 3-\mathrm{x} & & 3-\mathrm{x} & & 2\mathrm{x} & & \end{array}$$

$$\frac{\mathrm{x}^2}{(1-\mathrm{x}/2)(3-\mathrm{x}/2)} = \frac{4\mathrm{x}^2}{(3-\mathrm{x})^2} = \frac{4\mathrm{x}^2}{(2-\mathrm{x})(6-\mathrm{x})}$$

$$(3-\mathrm{x})^2 = (2-\mathrm{x})(6-\mathrm{x})$$

$$9+\mathrm{x}^2-6\mathrm{x} = 12-2\mathrm{x}-6\mathrm{x}+\mathrm{x}^2$$

$$9-6\mathrm{x} = 12-8\mathrm{x}$$

$$2\mathrm{x} = 3$$

$$\mathrm{x} = 1\cdot 5$$

$$\mathrm{K_c} = \frac{4\times 1\cdot 5\times 1\cdot 5}{1\cdot 5\times 1\cdot 5} = 4$$

Q.51 Answer:

$\frac{x}{y}+\frac{x^2}{2}=\lambda$

Solution:

Given equation is, $ydx - xdy + xy^2dx = 0$

Which could be converted into exact form

i.e. $\frac{ydx-xdy}{y^2} + xdx = 0$

$\Rightarrow \quad d\left(\frac{x}{y}\right) + d\left(\frac{x^2}{2}\right) = 0$

Integrating both the sides, we get $\frac{x}{y}+\frac{x^2}{2}=$ constant

or $\frac{x}{y}+\frac{x^2}{2}=\lambda$

Q.52 Answer:

$2009-\ln(2010-\ln 2010)$

Solution:

$\ln x = t$

$\frac{1}{x}dx = dt$

$x = e - 1t = 1$

$x = e^{2010} - 1t = \ln e^{2010}$

$= 2010$

$I = \int_1^{2010} 1 + \frac{1-t}{t(\ln x - \ln t)}dt$

$= \int_1^{2010} 1 + \frac{1-t}{t(t-\ln t)}dt$

$(t - \ln t) = y$

$\left(1 - \frac{1}{t}\right)dt = dy$

$\left(\frac{t-1}{t}\right)dt = dy$

$$= \underbrace{\int_1^{2010} 1dt}_{[2010-1]} + \underbrace{\int_1^{2010} \frac{1-t}{t(t-\ln t)}dt}_{\int \frac{-dy}{y}}$$

$= -\ln|y| = -\ln|t - \ln t]_1^{2010}$

$I = (2009) - \{\ln(2010 - \ln(2010)) - \ln(1-0)\}$

$I = 2009 - \ln(2010 - \ln 2010)$

Q.53 Answer:

$1 - \frac{2}{n}$

Solution:

Let total no. of persons $= n$

Since, two persons are selected

∴ total no. of ways of selection $= {}^nC_2$

Then $(n-2)$ **persons not selected are arranged in places shown below by circles and the selected 2 persons can be arranged at places stated by dots (Dots are** $n-1$ **in number) Therefore the favourable no. of ways** $= {}^{n-1}C_2$

$0 \cdot 0 \cdot 0 \cdot 0 \cdot 0 \cdot 0$

∴ P(**not together**) $= \frac{{}^{n-1}\mathrm{C}_2}{{}^n\mathrm{C}_2}$

$= \frac{(n-1)!}{2!(n-3)!} \times \frac{2!(n-2)!}{n!} = \frac{(n-2)(n-3)!(n-1)!}{(n-3)!n(n-1)!}$

$\mathrm{P} = \frac{(n-2)}{n} = 1 - \frac{2}{n}$

Q.54 Answer:

3

Solution:

$\lim_{x\to\infty} x^3 \ \ln\left(1 + \frac{1}{x}\right) + \frac{x}{2} - x^2$

$x = \frac{1}{t}$

$\lim_{t\to 0}\left(\frac{\ln(1+t)}{t^3} + \frac{1}{2t} - \frac{1}{t^2}\right) = \lim_{t\to 0} \frac{2\ \ln(1+t) + t^2 - 2t}{2t^3}$

$= \lim_{t\to 0} \frac{2\left(t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \ldots\right) + t^2 - 2t}{2t^3}$

$= \lim_{t \to 0} \left(\frac{1}{3} - \frac{t}{4} + \frac{t^2}{5} \ldots \ldots \right) = \frac{1}{3} = \frac{1}{m} \Rightarrow m = 3.$

Q.55 Answer:

$\frac{2I}{3}$

Solution:

Given $A + B = 2I \ldots (i)$

and $A^{-1} + B^{-1} = 3I \ldots (ii)$

Multiplying by matrix A in eq. (ii), we get, $AA^{-1} + AB^{-1} = 3AI$

$I + AB^{-1} = 3A$

Now, multiplying by matrix B, we get, $IB + AB^{-1}B = 3AB$

$\Rightarrow B + A = 3AB$

$\Rightarrow 3AB = 2I$(from (i))

$\Rightarrow AB = \frac{2I}{3}$

Q.56 Answer:

7. 26

Solution:

Let us assume an arbitrary mean $a = 155$ Following table is constructed:

x_i	f_i	$u_i = \frac{X_i - 155}{5}$	u_i^2	$f_i u_i$	$f_i u_i^2$
140	4	-3	9	-12	36
145	6	-2	4	-12	24
150	15	-1	1	-15	15
155	30	0	0	0	0
160	36	1	1	36	36
165	24	2	4	48	96
170	8	3	9	24	72
175	2	4	16	8	32
Total	125	-	-	77	311

$\therefore$ **Variance** $\sigma^2 = c^2 \left(\frac{\sum f_i u_i^2}{n} - \left(\frac{\sum f_i u_i}{n} \right)^2 \right) = 52.7076.$

Q.57 Answer:

99

Solution:

The given integral is $I = \int_0^{\tan 1} 0 dx + \int_{\tan 1}^{100\pi} 1 dx$

$\left(\text{as } \tan^{-1} x \in \left(1, \frac{\pi}{2}\right), \forall x > \tan 1 \right)$

$\Rightarrow I = 100\pi - \tan 1$

$= 100\pi + \tan(-1)$

$\therefore k = 100,\ p = -1$

$\therefore k + p = 99$

Q.58 Answer:

6

Solution:

Given,

$P(x) = x^6 - x^5 - x^3 - x^2 - x$

$P(x) = Q(x) \cdot \left(x^2 + 1\right) + \left(x^2 - x + 1\right)$

$P(\alpha) + P(\beta) + P(\gamma) + P(\delta) = \sum \alpha^2 - \sum \alpha + 4$

α, β, γ **and** δ **are the roots of** $Q(x) = 0$**. Thus, we get,**

$P(\alpha) + P(\beta) + P(\gamma) + P(\delta) = \left(\left(\sum \alpha\right)^2 - 2\sum \alpha\beta\right) - \sum \alpha + 4$

$P(\alpha) + P(\beta) + P(\gamma) + P(\delta) = (1 - 2(-1)) - 1 + 4$

$P(\alpha) + P(\beta) + P(\gamma) + P(\delta) = 6$

Q.59 Answer:

$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Solution:

$g(x) = f\left(\tan^2 x - 2\tan x + 4\right), 0 < x < \frac{\pi}{2}$

$g'(x) = f'\left(\tan^2 x - 2\tan x + 4\right) \times \left(2\tan x \sec^2 x - 2\sec^2 x\right)$

For $\mathrm{g(x)}$ **to be increasing**

$\mathrm{g'(x) > 0}$

$\mathrm{f'\left(\tan^2 x - 2\tan x + 4\right)\left(2\tan x \sec^2 x - 2\sec^2 x\right) > 0}$

$\mathrm{2\tan x \sec^2 x > 2\sec^2 x}$

$\mathrm{\tan x > 1}$

$\mathrm{\tan x > 1}$ **for** $\mathrm{x \epsilon \left(\frac{\pi}{4}, \frac{\pi}{2}\right)}$ **when** $\mathrm{\left[0 < x < \frac{\pi}{2}\right]}$

Q.60 Answer:

1

Solution:

As $\mathrm{f(x)}$ **is one-one function, so** $\mathrm{f\left(x^3 + 14x^2 + 13x - 5\right) = f\left(1 - x^2 + x^3\right)}$

$\Rightarrow x^3 + 14x^2 + 13x - 5 = 1x^3 + x^3 \Rightarrow 15x^2 + 13x - 6 = 0$

$\Rightarrow 15x^2 + 18x - 5x - 6 = 0 \Rightarrow (3x - 1)(5x + 6) = 0$

$\Rightarrow x = \frac{1}{3}, \frac{-6}{5}$ **but** $x \in [-1, 1]$

Hence $x = \frac{1}{3}$ **only .**

Q.61 Answer:

$\left(\frac{5}{4}, 1\right)$

Solution:

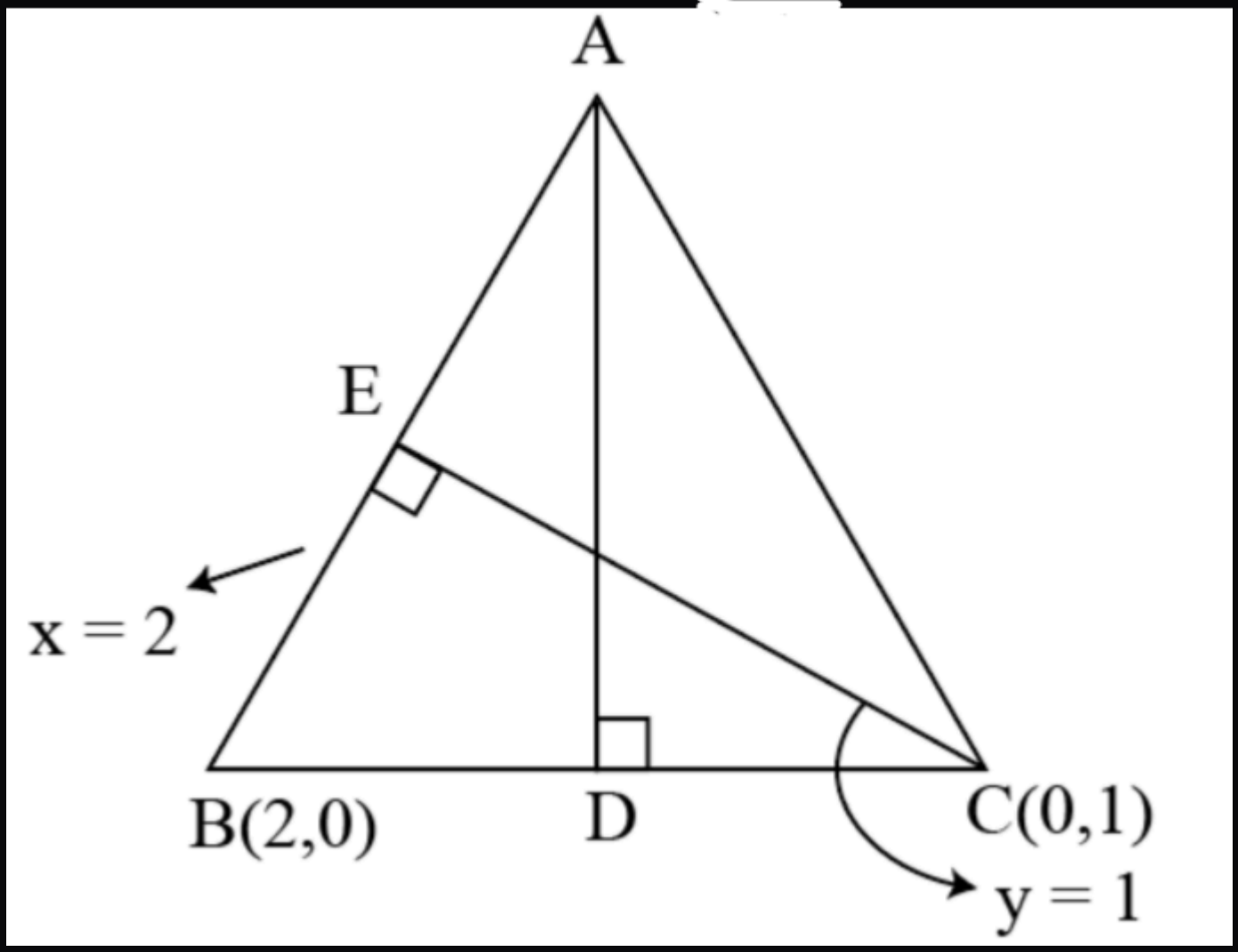
Since $AB = AC$

So, D is the midpoint of $BC \Rightarrow D = \left(1, \frac{1}{2}\right)$

$\Rightarrow Eq^n$ of AD is $\left(y - \frac{1}{2}\right) = -\frac{(0-2)}{(1-0)}(x-1)$

$\Rightarrow y - \frac{1}{2} = 2x - 2; \quad y = 2x - \frac{3}{2}$

Equation of CE is $y = 1 \quad \Rightarrow$ Ortho Centre is $\left(\frac{5}{4}, 1\right)$



Q.62 Answer:

$\frac{2r\pi}{n^2}$

Solution:

$(\cos\theta + i\sin\theta)(\cos 3\theta + i\sin 3\theta) \ldots\ldots = 1$

$e^{i(\theta)} \cdot e^{i(3\theta)} \ldots\ldots e^{i(2n-1)\theta} = 1$

$e^{i(\theta+3\theta+5\theta\ldots\ldots(2n-1)\theta)} = 1$

$e^{i\frac{n}{2}\{\theta+(2n-1)\theta\}} = 1$

$e^{in^2\theta} = 1$

$n^2\theta = 2r\pi$

$\theta = \frac{2r\pi}{n^2}$

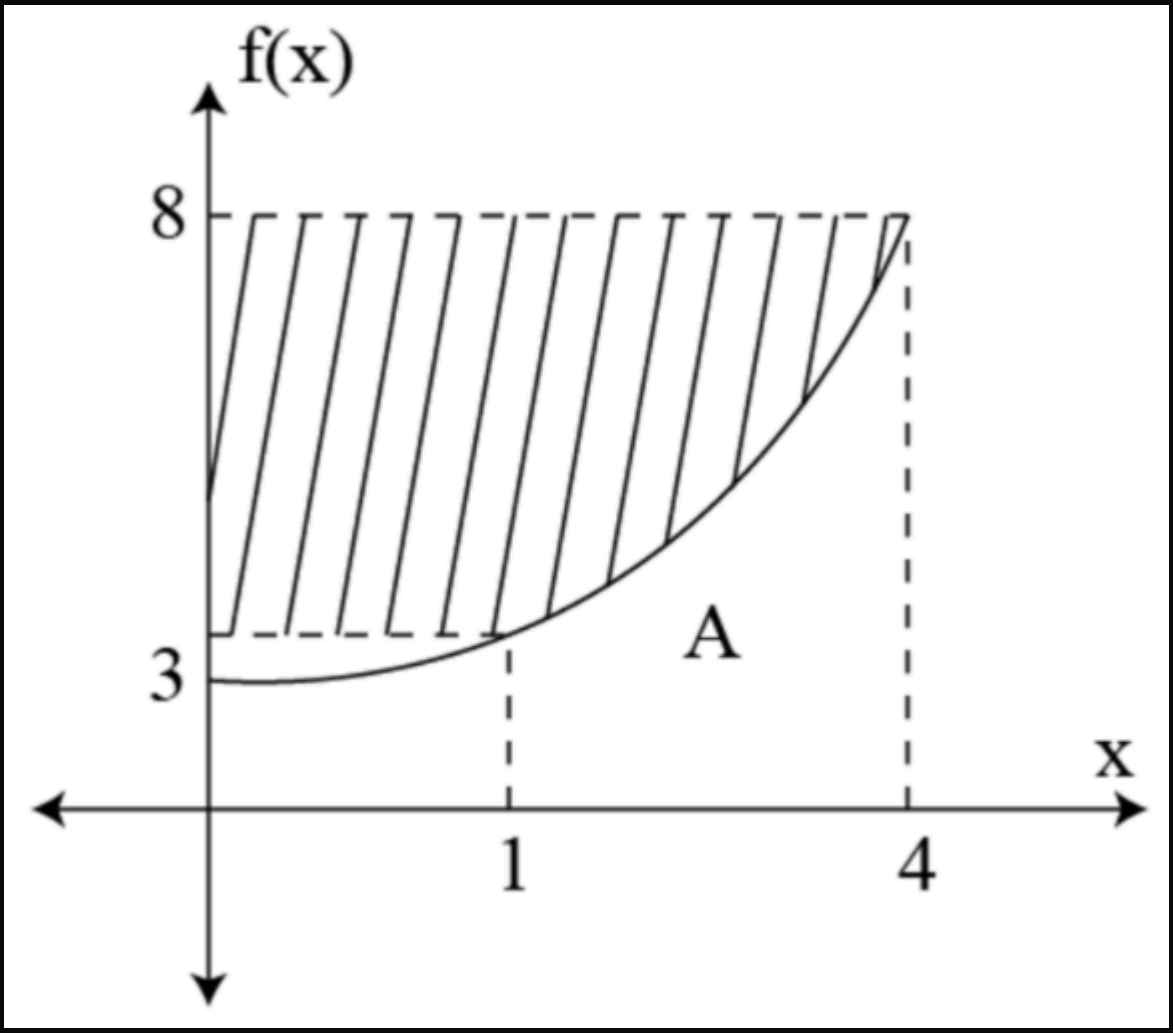
Q.63 Answer:

17

Solution:

Since $f(x) > 0 \forall x \in [1, 4]$

$\Rightarrow f^{-1}(x) > 0 \forall x \in [3, 8]$



Note that the shaded area is the required area and $A_1 = 12$.

Let the required area be A_2 Also, $A_1 + A_2 = 32 - 3 = 29$

$\Rightarrow A_2 = 17$ sq.units

Q.64 Answer:

3

Solution:

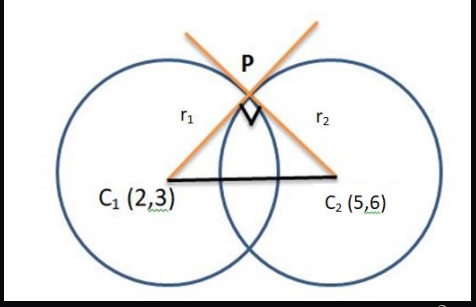
Given: Let $C_1 : (2, 3)$ *and* $C_2 : (5, 6)$ be the two given Centres of Circles and radius of both the circles are equal .

Calculations :

C_1C_2 be the distance between the centre of the circles .

$C_1C_2 = \sqrt{(5-2)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18}$

$\because$ Both Circles cut orthogonally { at 90° }.



Hence , Using the result $(C_1C_2)^2 = r_1^2 + r_2^2$, we get

$$\left(\sqrt{18}\right)^2 = r^2 + r^2 \qquad \{ \because \ r_1 = r_2 \ \}$$

$$\Rightarrow 2r^2 = 18$$

$$\Rightarrow r^2 = 9$$

$$\Rightarrow r = 3$$ **{ Since Radius is always positive }**

Q.65 Answer:

6

Solution:

We know that the number of subsets of a set containing n elements is 2^n.
Hence, the number of subsets of $A,\ B$ and C are respectively $2^\alpha,\ 2^\beta$ and 2^γ.
Given, $2^\alpha + 2^\beta + 2^\gamma = 28$
$\Rightarrow\ \alpha,\ \beta,\ \gamma$ **can be** $(2,\ 3,\ 4),\ (2,\ 4,\ 3),\ (3,\ 2,\ 4),\ (3,\ 4,\ 2),\ (4,\ 2,\ 3)$ **and** $(4,\ 3,\ 2)$
Hence, total 6 **possibilities are there.**

Q.66 Answer:

$$\frac{\left(\vec{a}\cdot\vec{c}\right)\vec{b}-\vec{c}\left(\vec{a}\cdot\vec{b}-1\right)}{\left(\vec{a}\cdot\vec{b}-1\right)}$$

Solution:

Given,

$$\left(\vec{a}\cdot\vec{x}\right)\vec{b} = \vec{c} + \vec{x} \ \ldots(1)$$

Taking dot product with $\vec{a}$**,**

$$\Rightarrow \left(\vec{a}\cdot\vec{x}\right)\left(\vec{b}\cdot\vec{a}\right) = \left(\vec{c}\cdot\vec{a}\right) + \left(\vec{a}\cdot\vec{x}\right)$$

$$\Rightarrow \left(\vec{a}\cdot\vec{x}\right)\left(\vec{b}\cdot\vec{a} - 1\right) = \left(\vec{c}\cdot\vec{a}\right)$$

$$\Rightarrow \left(\vec{a}\cdot\vec{x}\right) = \frac{\left(\vec{c}\cdot\vec{a}\right)}{\left(\vec{b}\cdot\vec{a}-1\right)} \ \ldots(2)$$

Substitute the value in (1)**,**

$$\left(\frac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}-1}\right)\vec{b} = \vec{c} + \vec{x}$$

$$\Rightarrow \vec{x} = \left(\frac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}-1}\right)\vec{b} - \vec{c}$$

$$\Rightarrow \vec{x} = \frac{\left(\vec{a}\cdot\vec{c}\right)\vec{b} - \vec{c}\left(\vec{a}\cdot\vec{b}-1\right)}{\left(\vec{a}\cdot\vec{b}-1\right)},\ \left(\vec{a}\cdot\vec{b} \neq 1\right)$$

Q.67 Answer:

0

Solution:

Given, $f(x) = \log_{10} x$ **and** $g(x) = e^{\pi i x}$, **where** $i = \sqrt{-1}$

Now, $f(x)g(x) = (\log_{10} x)\left(e^{\pi i x}\right)\ \ldots(\text{i})$

Also, $f(x)^{g(x)} = (\log_{10} x)^{\left(e^{\pi i x}\right)}\ \ldots(\text{ii})$

$$\therefore\ \phi(10) = \begin{vmatrix} (\log_{10} 10)\left(e^{\pi i(10)}\right) & (\log_{10} 10)^{\left(e^{\pi i(10)}\right)} & 1 \\ \left(\log_{10}(10)^2\right)\left(e^{\pi i(10)^2}\right) & \left(\log_{10}(10)^2\right)^{\left(e^{\pi i(10)^2}\right)} & 0 \\ \left(\log_{10}(10)^3\right)\left(e^{\pi i(10)^3}\right) & \left(\log_{10}(10)^3\right)^{\left(e^{\pi i(10)^3}\right)} & 1 \end{vmatrix}$$ **(from equations** (i) & (ii)**)**

$$\Rightarrow\ \phi(10) = \begin{vmatrix} e^{\pi i(10)} & 1 & 1 \\ (2)\left(e^{100\pi i}\right) & (2)^{\left(e^{100\pi i}\right)} & 0 \\ (3)\left(e^{1000\pi i}\right) & (3)^{\left(e^{1000\pi i}\right)} & 1 \end{vmatrix}\ [\because \log a^m = m\log a]$$

Also, $e^{10\pi i} = \cos(10\pi) + i\sin(10\pi) = 1,\ e^{100\pi i} = \cos(100\pi) + i\sin(100\pi) = 1$ **and**

$e^{1000\pi i} = \cos(1000\pi) + i\sin(1000\pi) = 1$

$$\therefore\ \phi(10) = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 0 \\ 3 & 3 & 1 \end{vmatrix}$$

Applying $R_3 \to R_3 - R_1$

$$\Rightarrow \phi(10) = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 0 \\ 2 & 2 & 0 \end{vmatrix} = 0$$ **(determinant with identical rows)**

Q.68 Answer:

6538

Solution:

Required number of ways = Coefficient of x^{10} **in** $\left(x^0 + x^1 + \ldots + x^5\right)^7$

= coefficient of x^{10} **in** $\left(1-x^6\right)^7(1-x)^{-7}$

= coefficient of x^{10} **in** $\left(1-7x^6\right)(1-x)^{-7}$ **(Neglecting the terms higher than** x^{10}**)**

= coefficient of x^{10} **in** $\left\{(1-x)^{-7} - 7x^6(1-x)^{-7}\right\}$

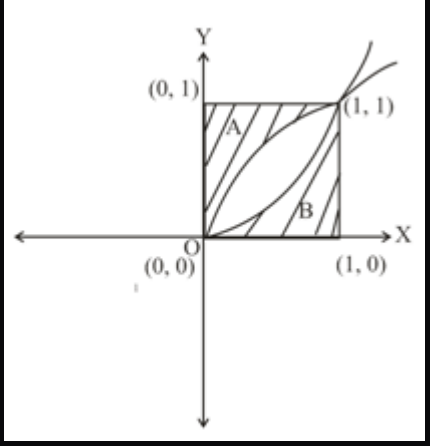
$= {}^{16}C_6 - 7.{}^{10}C_4 = 6538$

Q.69 Answer:

$P(A \cap B) = 0$

Solution:

Observe the figure below:



$P(A)$ and $P(B)$ do not have common area.

So $P(A \cap B\ )=0$

Q.70 Answer:

$\frac{x-2}{6}=\frac{y-4}{3}=\frac{z+1}{2}$

Solution:

Obtain the foot of the perpendicular $N(-4,\ 1,\ -3)$. Find the equations of the line passing through $P(2,\ 4,\ -1)$ and $N(-4,\ 1,\ -3)$

$\frac{x-2}{6}=\frac{y-4}{3}=\frac{z+1}{2}$

Q.71 Answer:

—

Solution:

It is given that,

$3f(x)+2f\left(\frac{x+59}{x-1}\right)=10x+30$

Putting $\mathrm{x}=7$ in eq.(1)

$$\Rightarrow 3\mathrm{f}(7)+2\mathrm{f}\left(\frac{7+59}{7-1}\right)=10\times 7+30$$

$$\Rightarrow 3\mathrm{f}(7)+2\mathrm{f}\left(\frac{66}{6}\right)=70+30$$

$$\Rightarrow 3\mathrm{f}(7)+2\mathrm{f}(11)=100\ldots\ldots\ldots(2)$$

Again putting $\mathrm{x}=11$ in eq.(1)

$$\Rightarrow 3\mathrm{f}(11)+2\mathrm{f}\left(\frac{11+59}{11-1}\right)=10\times 11+30$$

$$\Rightarrow 3\mathrm{f}(11)+2\mathrm{f}\left(\frac{70}{10}\right)=110+30$$

$$\Rightarrow 3\mathrm{f}(11)+2\mathrm{f}(7)=140$$

Multiplying eq(2) by 3 we get,

$$\Rightarrow 9\mathrm{f}(7)+6\mathrm{f}(11)=300$$

Multiplying eq(3) by 2 we get,

$$\Rightarrow 6\mathrm{f}(11)+4\mathrm{f}(7)=280\ldots\ldots\ldots(5)$$

Subtracting eq.(5) from eq.(4) we get,

$\Rightarrow [9f(7) + 6f(11)] - [6f(11) + 4f(7)] = 300 - 280$

$\Rightarrow 9f(7) + 6f(11) - 6f(11) - 4f(7) = 20$

$\Rightarrow 5\mathrm{f}(7) = 20$

$\Rightarrow f(7) = 4$

So, the required value is 4 .

Q.72 Answer:

—

Solution:

$|\mathrm{A}|^{2^4} = (2.5)^{16} \Rightarrow |\mathrm{A}| = \pm 10$

$\because |\mathrm{A}| = \mathrm{x} + \mathrm{y} + \mathrm{z}$, **where** $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathrm{N}$

$\therefore x + y + z = 10$

$\therefore$ **number of solutions** $= {}^{10-1}\mathrm{C}_{3-1} = {}^{9}\mathrm{C}_{2} = \frac{9\times 8}{2} = 36$

Q.73 Answer:

—

Solution:

$\frac{x+\frac{x^3}{3!}+\frac{x^5}{5!}+\cdots}{1+\frac{x^2}{2!}+\frac{x^4}{4!}+\ldots} = \frac{dx-dy}{dx+dy}$

$\frac{e^x}{e^{-x}} = \frac{(dx+dy)+(dx-dy)}{(dx+dy)-(dx-dy)}$

...[By Componendo Dividendo]

$\Leftrightarrow e^{2x} = \frac{dx}{dy}$

Integrating, we get

$y = \frac{-e^{-2x}}{2} + k$

$\Leftrightarrow 2ye^{2x} = 2ke^{2x} - 1$

$y(0) = 1 \Rightarrow k = \frac{3}{2}$

$\Rightarrow 2ye^{2x} = 3e^{2x} - 1$

$\Rightarrow f(x) = e^{2x}, m = 3, n = -1$

$\Rightarrow (m+n)f(0) = (3-1)(1) = 2$

Q.74 Answer:

—

Solution:

$10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$

Number of divisors $= 9 \times 5 \times 3 \times 2 = 270 = \lambda$

Q.75 Answer:

—

Solution:

$$\frac{1}{\sin 45 \sin 46} + \frac{1}{\sin 47 \sin 48} \ldots\ldots + \frac{1}{\sin 133 \sin 134}$$
$$= \frac{1}{\sin 45 \sin 46} + \frac{1}{\sin 46 \sin 47} + \ldots\ldots\ldots + \frac{1}{\sin 89 \sin 90}$$
Multiply and divide with $\sin 1$
$$= \frac{1}{\sin 1}\left[\frac{\sin(46-45)}{\sin 45 \sin 46} + \frac{\sin(47-46)}{\sin 46 \sin 47} + \ldots\ldots\ldots\ldots\right.$$
$$= \frac{1}{\sin 1}[\cot 45 - \cot 46 + \ldots\ldots + \cot 89 - \cot 90] = \frac{1}{\sin 1}$$